Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs, patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centered on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC)	999862809
Website(s)	www.ioba.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Paseo de Belén 17, 47011, Valladolid, España
Department(s), institute(s), research group(s) or unit(s) that would be involved	Retina Group, IOBA, University of Valladolid
Additional institutional resources relevant to collaboration	Facilities and equipment's of the IOBA-UVa (https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/) *and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Ivan Fernández Bueno, Associate Professor, Ophthalmology, UVa
Main contact email	Ivan.fernandez.bueno@uva.es
Other contact person(s), if relevant	Yolanda Álvarez, yolanda.alvarez.teruel@uva.es (Project Manager, Unidad de Excelencia IOBA)

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)

☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Biocompatibility of medical devices

Free Keywords for internal searchable database (please provide up to 10)	iPSC; retinal organoids; iRPE; animal models; preclinical models; neurodegeneration; biocompatibility; medical devices; neuroretina culture
Geographies of interest for exchanges (optional)	Germany; Ireland; New Zealand; Sweden

Overview of your research / innovation focus

Our research and innovation focus centres on advancing the understanding and treatment of inherited retinal dystrophies (IRDs) through a comprehensive preclinical approach. By investigating loss-of-function mechanisms associated with specific genetic mutations, we aim to clarify their phenotypic variability and underlying pathophysiology, acknowledging the complexity whereby different mutations may lead to similar clinical presentations and, conversely, the same mutation may produce divergent outcomes.

This work leverages patient-derived induced pluripotent stem cells (iPSCs), retinal organoids and gene-editing technologies such as CRISPR-Cas9 to correct mutations, model disease and dissect molecular mechanisms. Current efforts include the in vitro and in vivo characterisation of IRDs linked to the c.1354dupT mutation in the PROM1 gene, using iRPE models and knock-in mouse systems, with a strong emphasis on rare diseases and translational applicability.

In parallel, we explore neuroprotective strategies aimed at preserving retinal structure and function, including repurposing existing drugs to mitigate neurodegeneration, while also contributing to the preclinical biocompatibility assessment of ophthalmic medical devices under ISO 10993 standards. This combined approach supports the safe and effective translation of novel therapies and devices into clinical practice through the evaluation of cytotoxicity, irritation, sensitisation and toxicity.

Complementing these experimental and translational activities, the team currently co-leads the national multi-centre project StargSpain (Precision Medicine and Advanced Diagnosis in Stargardt Disease in Spain), focused on the phenotypic and genotypic characterisation of patients with ABCA4-associated retinopathies using whole genome sequencing and multimodal retinal imaging. The project also incorporates Artificial Intelligence and Deep Learning approaches to develop advanced diagnostic and predictive tools based on OCT and fundus autofluorescence, strengthening the integration of clinical datasets with our preclinical research lines..

Why your team could be relevant to a translational eye-health and vision consortium

Our team brings distinctive added value to a consortium focused on visual health and translational ophthalmology through an integrated, end-to-end preclinical capability that bridges fundamental research and clinical application. We contribute deep expertise in modelling inherited retinal dystrophies using patient-derived iPSC systems and advanced gene-editing tools such as CRISPR-Cas9, enabling precise investigation of disease mechanisms and validation of therapeutic strategies in a human-relevant context. Our ongoing work on PROM1-associated retinal degeneration, combining in vitro retinal organoids and iRPE models with in vivo knock-in mouse systems, positions us to generate robust, translatable data for rare and genetically complex conditions.

In parallel, our research on neuroprotection provides a complementary therapeutic dimension, focused on preserving retinal structure and function across degenerative diseases, including through drug-repurposing approaches that can accelerate clinical translation. This dual focus on disease modelling and therapeutic intervention strengthens the consortium's capacity to move from mechanistic insight to actionable treatment strategies.

Additionally, our dedicated platform for preclinical biocompatibility assessment of ophthalmic drugs and medical devices, aligned with ISO 10993 standards, offers critical infrastructure for safety evaluation. This enables early identification of cytotoxicity, irritation, sensitisation and systemic toxicity risks, facilitating regulatory readiness and de-risking innovation pipelines.

Our team also contributes expertise in multimodal retinal imaging, genotype–phenotype correlation studies, and the application of Artificial Intelligence and Deep Learning tools for advanced diagnosis and prediction in inherited retinal diseases. This clinical and computational dimension reinforces our ability to integrate robust patient-oriented datasets with preclinical research, supporting precision medicine strategies and clinically meaningful translational outcomes. Altogether, our team enhances the consortium's ability to deliver scientifically rigorous, clinically relevant and translationally impactful outcomes.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- **Advanced therapy development pipeline** spanning neuroprotection and regenerative medicine approaches for retinal diseases, with a strong orientation toward clinically actionable outcomes, supported by extensive experience in preclinical development and validation, as well as clinical translation through investigator-initiated trials in cell-based therapies for retinal disorders, and gene therapy strategies leveraging CRISPR-Cas technologies.
- **Organotypic neural retina culture platform** enabling ex vivo evaluation of neuroprotective compounds, providing a physiologically relevant intermediate between cell culture and in vivo models.
- **Custom preclinical model generation** (both in vitro and in vivo) tailored to specific retinal pathologies, supporting efficacy testing and mechanism-of-action studies for emerging therapies.
- **Expertise in iPSC technology and retinal organoids**, allowing the study of human disease biology in complex, patient-specific systems and facilitating therapeutic screening.
- **Multimodal retinal imaging and genotype–phenotype correlation approaches,** integrating whole genome sequencing (WGS), OCT, fundus autofluorescence and Artificial Intelligence/Deep Learning methodologies for advanced diagnosis, prediction and precision medicine strategies in inherited retinal diseases.
- **Integrated innovation ecosystem**, including leadership in competitive research projects, active collaboration with industry partners, and co-invention of patents that drive technology transfer and real-world application.
- **Translational acceleration capacity**, combining experimental validation with strategic alignment toward regulatory and clinical development pathways.
- **Specialised platform for ophthalmic device safety assessment**, supporting early-stage validation and risk mitigation in parallel with therapeutic development.
- **Training and knowledge transfer hub**, with the ability to host and train visiting researchers in advanced retinal models and techniques, as well as to send team members abroad to acquire and implement cutting-edge methodologies.
- **Collaborative flexibility**, adapting experimental systems and expertise to consortium needs, from mechanistic studies to preclinical proof-of-concept and technology validation.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Facilities and equipment's of the IOBA-UVa (*https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/*):

- *Cell Culture Lab*: equipped with laminar flow hoods, incubators, ovens, water baths, stereo, light and epifluorescence microscopes, electroporator, cell counters, transfectors, centrifuges, liquid nitrogen tanks, -80°C and -20°C freezers, among others.
- *Ocular Pathology Lab*: equipped with automated tissue processors, paraffin embedding unit, microtomes, cryostat, routine staining batteries, and epifluorescence microscope.
- *Molecular Biology Lab*: equipped with PCR and RT-PCR equipment, laminar flow hood, centrifuges, GenePix, microplate reader, gel electrophoresis system, and sonicator, anmong others.
- *Shared Labs*: equipped with autoclaves, laminar flow hoods, ovens, precision balances, pipettes, ice machines, sterilization equipment, centrifuges, and pHmeters, among others.
- *Preclinical and translational platforms:* established infrastructure for ex vivo organotypic neural retina culture, in vitro retinal disease models, and in vivo animal models for efficacy, safety and neuroprotection studies.
- *Clinical and multimodal imaging datasets:* access to phenotypic, genotypic and multimodal retinal imaging datasets associated with inherited retinal diseases, including OCT and fundus autofluorescence data integrated with whole genome sequencing and Artificial Intelligence/Deep Learning analytical approaches.

Through the centralized services of Uva, access is available to:

- *Instrumental Techniques Laboratory (LTI-INFRARED):* for confocal microscopy and flow cytometry studies (LTI-INFRARED Platform, https://aplicaciones.educa.jcyl.es/infrared/welcome.action).
- *Animal Research and Welfare Service (SIBA):* providing specialised facilities for housing, breeding, monitoring and experimental procedures involving animal models.

Training capacity and knowledge-transfer potential

Inbound staff in our organisation would benefit from a highly interdisciplinary training environment, with a primary focus on scientific, technical, clinical, computational and translational skills related to retinal diseases and visual health research. They would gain hands-on experience in advanced retinal disease modelling, including iPSC-derived systems, retinal organoids, iRPE models, and the development and application of tailored in vitro and in vivo preclinical models.

From a technical perspective, visitors would be trained in cutting-edge methodologies such as gene editing, including CRISPR-Cas-based approaches, as well as experimental design for the evaluation of novel therapeutic strategies. A key added value is the opportunity to test and validate new compounds—either developed within their home institutions or generated through collaborative projects—using our established preclinical platforms, particularly our ex vivo organotypic neural retina culture model, in which we have extensive expertise and a strong track record of robust and reproducible neuroprotection studies. Training would also include preparation, maintenance and application of these experimental systems, together with downstream cellular, molecular and functional analyses.

In addition, inbound researchers could gain exposure to multimodal retinal imaging analyses, genotype–phenotype correlation studies, and the integration of Artificial Intelligence and Deep Learning tools applied to inherited retinal diseases, including OCT and fundus autofluorescence-based approaches linked to precision medicine strategies.

On the translational side, staff would gain insight into how preclinical and clinical-oriented findings are structured and validated to support downstream application, including alignment with clinically meaningful endpoints and regulatory considerations. Overall, the training environment provides a comprehensive framework for advancing innovative therapeutic and diagnostic approaches in retinal diseases from early-stage research towards translational impact.

What your team would hope to gain from participating

Our team aims to gain access to complementary expertise, advanced technologies and innovative translational approaches that can strengthen and expand our current research lines in retinal diseases and visual health. In particular, we are interested in establishing collaborations with groups working on novel neuroprotective compounds, advanced gene and cell therapy strategies, multimodal imaging, Artificial Intelligence applications, and emerging analytical platforms that could enhance the depth and translational relevance of our experimental findings.

A central objective is to build long-term international scientific collaborations that extend beyond short-term exchanges and evolve into sustainable partnerships supported by joint projects, coordinated funding applications, scientific communications and high-impact publications. We are especially interested in integrating complementary datasets, technologies and methodological approaches that improve the robustness, reproducibility and clinical extrapolation of preclinical and patient-oriented results.

We strongly support the mobility and international training of PhD students, who would benefit from exposure to new research environments, experimental methodologies, institutional cultures and multidisciplinary working dynamics. These experiences are expected to significantly enrich their scientific and professional development, fostering adaptability, independence, international networking and broader perspectives on translational research.

At the same time, senior researchers and technical staff participating in exchanges would have the opportunity to acquire and implement novel methodologies, analytical tools and experimental strategies that could be strategically incorporated into our laboratories and research workflows. This includes exposure to infrastructures and technologies not currently available in our institution, as well as the possibility of refining key translational and methodological aspects to ensure more reliable and clinically meaningful interpretation of results.

Overall, participation in the consortium is expected to promote a dynamic and bidirectional transfer of knowledge, strengthening scientific excellence, innovation capacity and long-term collaborative impact across all participating groups.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Iván Fernández Bueno	Associate Professor and Principal Investigator	IOBA	Researcher	Experienced	ivan.fernandez.bueno@uva.es
Rosa Coco Martín	Full Professor and Principal Investigator	IOBA	Researcher	Experienced	rosamcoco@uva.es
Yenisey García Ferrer	PhD candidate	IOBA	Researcher	Early-Stage	yenisey.garcia22@uva.es
Yolanda Álvarez Teruel	Project Manager	IOBA	Managerial	Experienced	Yolanda.alvarez.teruel@uva.es
TBD	2 x Predoctoral Students	IOBA	Researcher	Early-Stage	

Main disciplines represented in your team	Cell Biology, Molecular Biology, Genetics and Genomics, Experimental and Translational Ophthalmology, Retinal Disease Modelling, Regenerative Medicine, Neuroprotection, Gene Therapy, Preclinical and Bioengineering Models, Artificial Intelligence and Deep Learning applied to retinal imaging, Precision Medicine, and Multimodal Retinal Imaging Analysis.
Experience in EU / international / multi-partner collaborative projects	Participation as an early-stage researcher in a secondment within the *Drug Discovery and Development of Novel Eye Therapeutics (3D-NET)* project (FP7-PEOPLE-2013-IAPP), where I also designed, conducted, and led experimental work supervising seconded staff at my home institution (UVA/IOBA).
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | **Short note:** *Translational/preclinical retinal research; no direct patient-facing clinical activity indicated in this profile.*

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. *Puertas-Neyra K, Coco-Martin RM, Hernandez-Rodriguez LA, Gobelli D, Garcia-Ferrer Y, Palma-Vecino R, Tellería JJ, Simarro M, de la Fuente MA, Fernandez-Bueno I.* ***Clinical exome analysis and targeted gene repair of the c.1354dupT variant in iPSC lines from patients with PROM1-related retinopathies exhibiting diverse phenotypes****. Stem Cell Research & Therapy 2024;15:192. DOI: 10.1186/s13287-024-03804-2*
2. *Usategui-Martín R, Puertas-Neyra K, Galindo-Cabello N, Hernández-Rodríguez LA, González-Pérez F, Rodríguez-Cabello JC, González-Sarmiento R, Pastor JC, Fernandez-Bueno I*.* ***Retinal Neuroprotective Effect of Mesenchymal Stem Cells Secretome Through Modulation of Oxidative Stress, Autophagy, and Programmed Cell Death****. Investigative Ophthalmology & Visual Sciences 2022;63:27. DOI: 10.1167/iovs.63.4.27*
3. *Usategui-Martín R, Puertas-Neyra K, García-Gutierrez MT, Fuentes M, Pastor JC, Fernandez-Bueno I*.* ***Human mesenchymal stem cell secretome exhibits a neuroprotective effect over in vitro retinal photoreceptor degeneration****. Molecular Therapy - Methods & Clinical Development 2020;17:1155-1166. DOI: 10.1016/j.omtm.2020.05.003*
4. *Srivastava GK, Alonso-Alonso ML, Fernandez-Bueno I, Garcia-Gutierrez MT, Rull F, Medina J, Coco RM, Pastor JC.* ***Comparison between direct contact and extract exposure methods for PFO cytotoxicity evaluation****. Scientific Reports 2018;8:1425. DOI: 10.1038/s41598-018-19428-5*
5. *Fernandez-Bueno I, Fernandez-Sanchez L, Gayoso MJ, Garcia-Gutierrez MT, Pastor JC, Cuenca N.* ***Time course modifications in organotypic culture of human neuroretina****. Experimental Eye Research 2012; 104:26-38. DOI: 10.1016/j.exer.2012.08.012*

Up to 5 relevant previous projects, grants, networks or collaborations

Design and characterization of an in vitro model of Retinitis Pigmentosa due to a mutarion on the PROM1 gene by using genetic edition of induced Pluripotent Stem Cells (RP-PROM1)
Diseño y caracterización de un modelo in vitro de Retinitis Pigmentosa por mutación en el gen PROM1 mediante edición genética de Células Madre Pluripotenciales inducidas (RP-PROM1)

- *Proyectos de I+D+i – Generación de Conocimiento, Convocatoria 2020, Ministerio de Ciencia e Innovación.*
- *Referencia: PID2020-118860RB-I00*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid.*
- *Coco Martín RM (Investigador Principal 1), Fernández Bueno I (Investigador Principal 2), Srivastava GK (Equipo de Trabajo), Tellería Orriols JJ (Equipo de Trabajo), García Gutiérrez MT (Equipo de Trabajo).*
- *Importe: 105.335€*
- *4 años, 01/09/2021-31/08/2025.*

Design, Development and Validation of a Highly Purified Perfluorooctane for Use in Ophthalmic Surgery: JUSTPERFLUORO
Diseño, desarrollo y validación de un perfluroooctano altamente purificado para uso en cirugía oftalmológica: JUSTPERFLUORO

- *Proyectos de colaboración público-privada 2021, Ministerio de Ciencia e Innovación.*
- *Referencia: CPP2021-008695*
- *AJL Opththalmic S.A.; VISION I MAS D S.L.; y Universidad de Valladolid (Instituto de Oftalmobiología Aplicada -IOBA- y Grupo de Tecnologías a Presión -PressTech-)*
- *AJL Opththalmic S.A.: Larra E (Coordinadora), Barrio P (Investigador), Cobo MT (Investigador), Uriarte N (Investigador), Martínez A (Investigador), Gondra Vázquez A (Investigador), Peña Aldrey CA (Investigador); VISION I MAS D S.L.: García Gutiérrez MT (Investigador); Universidad de Valladolid: Fernández Bueno I (Investigador Principal), García Serna J (Investigador), Cantero Sposetti DA (Investigador) y Pastor Jimeno JC (Investigador).*
- *Importe: 925.208€ (AJL Opththalmic S.A.: 194.974,11€; VISION I MAS D S.L.: 147.978,60€; Universidad de Valladolid: 110.784,00€)*
- *3 años, 01/03/2022-31/12/2025.*

In Vitro and In vivo Modeling and characterization of Inherited Retinal Diseases due to the c.1354dupT mutation in PROM1, using iRPE and murine knock-in models (IVIM-PROM1).
Modelado y caracterización in vitro y in vivo de las Distrofias Hereditarias de la Retina originadas por la mutación c.1354dupT en PROM1, mediante modelos iRPE y ratones knock-in (IVIM-PROM1)

- *Proyectos de Generación de Conocimiento, Convocatoria 2024, Ministerio de Ciencia, Innovación y Universidades*

- *Referencia: PID2024-160341OB-I00*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid.*
- *Fernández Bueno I (Investigador Principal 1), Coco Martín RM (Investigador Principal 2), Tellería Orriols JJ (Equipo de Investigación) , Cuadrado Asensio R (Equipo de Investigación), Puertas Neyra KL (Equipo de Trabajo), Hernández Rodríguez LA (Equipo de Trabajo), Andrés Iglesias C (Equipo de Trabajo), Bravo Nuevo A (Equipo de Trabajo), García Vázquez C (Equipo de Trabajo)*
- *Importe: 141.250€*
- *4 años, 01/09/2025-31/08/2029.*

Ocular Topical administration of Sorbitan ester Nanoparticles for Intraocular Neuroprotection (RETINARTEX)
Administración tópica ocular de nanopartículas de ésteres de sorbitán para neuroprotección intraocular (RETINARTEX)

- *Ayudas a Proyectos de "Prueba de Concepto" 2022, Ministerio de Ciencia e Innovación*
- *Referencia: PDC2022-133239-C21*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid*
- *Diebold Luque Y (Investigador Principal 1), García Posadas L (IP 2), Fernández Bueno I (Investigador Colaborador), Puertas Neyra KL (IC), Romero Castillo I (IC), Ossa Calderón C (IC), Andrés Iglesias C (IC), Sánchez Barreiro A (IC).*
- *Importe: 115.000€*
- *2 años, 01/12/2022-30/11/2024*

Precision Medicine and Advanced Diagnosis in Stargardt Disease in Spain: Diagnostic Tools through the Use of Genetic Biomarkers, Imaging, and Artificial Intelligence Analysis. Acronym: StarSpain
Medicina de precisión y diagnóstico avanzado en la Enfermedad de Stargardt en España: Herramientas Diagnósticas mediante el Uso de Biomarcadores Genéticos, de Imagen y Análisis con Inteligencia Artificial (StargSpain)

- *Proyectos de Investigación en Enfermedades Raras 2024, Misiones Conjuntas del Ministerio de Sanidad y del Ministerio de Ciencia, Innovación y Universidades, Instituto de Salud Carlos III*
- *Referencia: PMPER24/00022*
- *Fundación Instituto de Investigación Sanitaria de Aragón, Instituto de Investigación Sanitaria de Aragón, Universidad de Zaragoza*
- *Grupos Participantes: Instituto de Investigación Sanitaria de Aragón, Universidad de Valladolid, Hospital de la Paz, Biodonostia, Instituto oftalmológico Fernández-Vega, Hospital Virgen Macarena, Universidad de Barcelona, Hospital La Fé, Universidad de Navarra*
- *Pinilla Lozano I (IP), Coco Martín RM (IP Corresponsable), Fernández Bueno I (Investigador Colaborador UVa), Sánchez Tocino H (IC UVa), Cuadrado Asensio R (IC UVa), Andrés Iglesias C (IC UVa), Blázquez Araúzo F (IC UVa)*
- *Importe: 1.167.650€ (UVa: 162.341,69€)*
- *2 años, 01/01/2025-31/12/2026*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Pharmaceutical Composition for the Treatment and/or Prevention of Degenerative Retinal Diseases
Composición farmacéutica para el tratamiento y/o prevención de enfermedades retinianas degenerativas

- *Inventores: Srivastava GK, Singh AK, Fernández Bueno I, Gayoso MJ, Pastor Jimeno JC*
- *Número de patente o CCP: ES 2539329 A1*
- *Fecha de concesión: 29/06/2015*
- *Número de la solicitud: P201331743*
- *Fecha de solicitud: 29/11/2013*
- *Número de patente internacional: WO/2015/079093*
- *Fecha de concesión: 04/06/2015*
- *Número de la solicitud internacional: PCT/ES2014/070879*
- *Clasificación Internacional de Patentes: C12N 5/0775 (2010.01)*
- *País de prioridad: España*
- *Entidad titular: Universidad de Valladolid*

Method for Evaluating the Cytotoxicity of Chemical Substances
Método para evaluar la citotoxicidad de sustancias químicas

- *Inventores: Srivastava GK, Alonso Alonso ML, Fernández Bueno I, García Gutiérrez MT, Coco Martín RM, Pastor Jimeno, JC.*
- *Número de patente o CCP: ES2644987*
- *Número de patente europea: EP2080.12-PACE*
- *Fecha de concesión: 05/10/2018*
- *Número de la solicitud: P201630708*
- *Fecha de solicitud: 30/05/2016*
- *Número de la solicitud internacional: PCT/ES2017/070365*
- *Fecha de solicitud: 29/05/2017*
- *Número de publicación: EP3467118*
- *Fecha de publicación: 10/04/2019*
- *País de prioridad: España*
- *Entidad titular: Universidad de Valladolid*

- *Validación externa: "External validation of a medical devices cytotoxicity testing method: IOBA-CYTOTEST". Final Report Study Code: 4465 14 november 2019*
- *Empresa/s que la están explotando: Vision I+D S.L., Valladolid, España*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation X Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	None

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

- **Organotypic neural retina culture platform** as a core strength for the **direct evaluation of novel molecules and products**, combining hands-on experimental workflows with robust readouts through cellular and molecular biology techniques, as well as functional assays; including full training in the preparation, handling, and application of this ex vivo model.
- **Advanced human retinal modelling using iPSC technology**, including reprogramming, differentiation into retinal cell types, and generation of retinal organoids and iRPE systems for disease modelling and therapeutic testing.
- **In vivo preclinical models of retinal disease**, enabling the modelling and characterisation of pathologies and the evaluation of new therapeutic approaches in a physiologically relevant context.

Capabilities or expertise you would especially like to access through the consortium

- **Discovery and validation of neuroprotective compounds** through collaboration with complementary teams (academic and industrial), enabling the identification of novel candidates with potential relevance for retinal diseases and their evaluation across diverse preclinical platforms.
- **Advanced gene therapy approaches in vivo**, particularly partnerships that enable the implementation and optimisation of CRISPR-Cas-based strategies in animal models, with a clear focus on strengthening the pathway toward clinical translation in human retinal disorders.
- **Access to cutting-edge infrastructures and specialised technologies** not currently available in our centre, including advanced imaging, omics platforms, high-throughput screening, AI-driven analytical tools and large-scale computational approaches that could maximise the value and translational relevance of our datasets and experimental results.
- **Integration of multimodal clinical and imaging datasets**, including expertise in Artificial Intelligence, Deep Learning and genotype–phenotype correlation analyses to strengthen precision medicine approaches and predictive modelling in inherited retinal diseases.
- **Improved translational and methodological standardisation**, refining key experimental and analytical aspects that facilitate robust, reproducible and clinically meaningful extrapolation of preclinical findings.
- **Effective communication and dissemination strategies**, aiming not only to reach the scientific community through high-impact publications, but also to improve outreach to society, increasing awareness and understanding of retinal diseases and emerging therapies.
- **Long-term international collaboration frameworks**, fostering durable partnerships supported by joint research projects, coordinated funding applications, and co-authored scientific outputs.
- **Exchange of expertise and methodologies**, enabling mutual learning and integration of innovative techniques that enhance experimental capabilities across participating groups.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Ivan Fernández Bueno
Date completed: 18/5/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

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Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Valladolid
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	Main institutional website and any relevant team / laboratory / company webpages
Country	Spain
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Street, city, postcode, country Campus Miguel Delibes. Paseo de Belén, 17. 47011 Valladolid, Spain
Department(s), institute(s), research group(s) or unit(s) that would be involved	Please name the unit(s) most relevant to the potential MSCA-SE action Ocular Surface Research Group, Institute of Applied Ophthalmobiology (IOBA), University of Valladolid.
Additional institutional resources relevant to collaboration	e.g. core facilities, clinical units, technology platforms, animal facilities, data infrastructures Facilities and equipment's of the IOBA-UVa (https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/) *and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Name, title, position, department / unit Margarita Calonge, Full Professor of Ophthalmology, Director of IOBA, UVa.
Main contact email	mcalonge@uva.es
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☒ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☒ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☒ Advanced therapies / regenerative medicine	☒ Gene therapy / cell therapy / tissue engineering
☒ Inflammation / biomarkers / neuroprotection	☒ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☒ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☒ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	e.g. retinal organoids; ocular surface inflammation; zebrafish; AI diagnostics; drug delivery Ocular surface inflammation; cell therapy; regenerative medicine; ocular pain; mesenchymal stem cells; clinical trials; secretome; extracellular vesicles; nanomedicine
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus

Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words).

Our team focuses on the translational research and development of advanced therapies for severe inflammatory ocular surface diseases (OSD), particularly limbal stem cell deficiency (LSCD), severe dry eye disease (i.e. Sjögren syndrome, ocular graft-versus-host disease (oGVHD)), and also as novel therapy for corneal neuropathic pain. The group combines expertise in ophthalmology, stem cell biology, regenerative medicine, immunology and nanobiotechnology to develop innovative therapeutic strategies to treat OSD and reduce vision loss.

Our current research is mainly focused on mesenchymal stromal/stem cell (MSC)-based therapies, including the development of genetically modified and nanoengineered MSCs with enhanced migratory, immunomodulatory and regenerative properties. In collaboration with expert groups in nanomedicine, we are developing cell-based and cell-free therapeutic approaches using nano-engineered MSCs as targeted drug delivery systems for ocular surface inflammation.

Our work integrates *in vitro* inflammatory models, preclinical animal models and translational studies with strong focus on clinical application (having successfully translated some therapies to the clinical field). The team has extensive experience in advanced therapy medicinal products (ATMPs), ocular surface regeneration, subconjunctival cell delivery and translational ophthalmology, and actively collaborates with clinical, industrial and regulatory partners.

The group is particularly interested in interdisciplinary collaborations involving regenerative medicine, nanotechnology, gene and cell engineering, immunomodulation, epithelial-immune interactions and cell-free therapeutic strategies for inflammatory OSD and neuropathic corneal pain.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our team could bring strong added value to a translational eye-health and vision consortium through its multidisciplinary expertise in regenerative medicine, stem cell biology, nanotechnology and translational ophthalmology. We have extensive experience in the development of advanced therapy medicinal products (ATMPs) for ocular surface diseases, integrating *in vitro* models, clinically relevant preclinical models and translational approaches with direct clinical applicability.

A key contribution of our group is the development of innovative mesenchymal stromal/stem cell (MSC)-based therapies, including genetically modified and nanoengineered MSCs with enhanced immunomodulatory and regenerative properties. In collaboration with experts in nanomedicine, we have developed cell-based targeted drug-delivery strategies and are currently exploring cell-free therapeutic approaches using MSC-derived secretome. These technologies may be highly relevant not only for ophthalmology, but also for a broader range of inflammatory and degenerative diseases, opening opportunities for cross-disciplinary innovation.

The group also possesses expertise in subconjunctival administration as a minimally invasive delivery route, ocular surface inflammation, epithelial-immune interactions and the translational validation of regenerative therapies. Importantly, our experience extents the full translational pathway, from experimental development and preclinical validation to clinical application and collaboration with industrial partners. The group also have clinicians expert in the diseases.

We believe this combination of ophthalmology, regenerative medicine, nanotechnology and translational research could contribute to generating innovative and clinically relevant solutions within a highly interdisciplinary consortium.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.*

- Development of advanced therapy medicinal products (ATMPs) for ocular surface diseases.
- Mesenchymal stromal/stem cell (MSC) biology, expansion and characterisation from adipose tissue and bone marrow sources.
- Genetic modification and nanoengineering of MSCs to enhance migratory, immunomodulatory and regenerative properties.
- Development of MSC-derived secretome (extracellular vesicles approaches for inflammatory disease modulation.
- *In vitro* models of ocular surface inflammation and epithelial-immune interactions.
- Multiparametric flow cytometry and spectral cytometry for immunophenotyping and receptor analysis.
- Multiplex cytokine and inflammatory biomarker profiling (Luminex technology, ELISA).
- Clinically relevant rabbit models of limbal stem cell deficiency (LSCD) and ocular surface regeneration.
- Expertise in minimally invasive subconjunctival cell administration.
- Histology, immunofluorescence and tissue characterisation of the ocular surface.
- Translational ophthalmology, including preclinical validation and clinical implementation of cell-based therapies.
- Experience in investigator-driven clinical trials, compassionate-use therapies and interaction with Spanish Agency of Medicines and Medical Devices (AEMPS).
- Collaboration with GMP-compliant cell production facilities, CROs and industrial partners.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities

.

-Two Cell culture laboratories: equipped with three laminar flow hoods, three CO_2 incubators, two thermostatic bath, two inverted phase contrast microscopes, and two centrifuges.

- Experimental ocular pathology laboratory: equipped with everything necessary for the processing and analysis of histological samples for light microscopy (regular and special stains, immunohistochemistry, etc.).
- Fluorescence microscope with image capture and treatment system (Leica DM6000B, Wtzlar, Germany).
- Fluorescence automated upright microscope DM6B THUNDER (Leica)
- Access to a Leica TC SP8 Lightning confocal microscope at the Advanced Optical Microscopy Service Instrumental Techniques Laboratory of the University of Valladolid.
- SpectraMAx M5 spectrophotometer (Molecular Devices).
- Real time PCR system PCR 7500 System (Applied Biosystems).
- Heidelberg Retina Tomograph III in combination with the Rostock Cornea Module: (IOBA-UVa)
- Swept-source optical coherence tomography technology (Plex Elite; Carl Zeiss Meditec): (IOBA-UVa)
- Ocular Surface and Immunology Clinical Unit: this is the clinical unit of IOBA where ocular surface disease patients, are diagnosed and treated.
- OculoFacial Pain Clinical Unit: this is the clinical unit at IOBA where patients with severe ocular chronic pain are diagnosed and treated

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Inbound researchers joining our research group will receive multidisciplinary training in the field of ocular surface regenerative medicine and advanced cell therapy. The host institution offers a highly translational environment integrating basic research, GMP-oriented manufacturing approaches, preclinical development and clinical application. Researchers will have the opportunity to acquire and strengthen the following competencies:

Scientific and technical skills:

Training in mesenchymal stem cell and epithelial cell culture, ex vivo models, nanoengineering, microscopy, molecular and cellular biology techniques, immunohistochemistry, and experimental in vivo models of ocular surface diseases.

Clinical and translational skills:

The ophthalmologists of this group have wide experience in severe diseases affecting the ocular surface and have already participated in the translational aspects of the research, already treating patients with stem cell therapy.

Regulatory and quality-management skills:

Introduction to European regulatory frameworks governing Advanced Therapy Medicinal Products (ATMPs), ethical requirements, quality assurance and GMP-related standards relevant to cell-based therapies.

Data analysis and digital skills:

Image analysis, experimental design, data reproducibility and management of biomedical research datasets.

Transferable and managerial skills:

Development of scientific communication abilities through manuscript preparation, conference presentations and interdisciplinary teamwork.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Participation in the programme would provide significant opportunities to strengthen the scientific, translational and collaborative capacity of our research group beyond our current expertise in ocular surface cell therapy.

We expect to benefit from the acquisition of advanced experimental approaches, bioengineering strategies, inflammation *in vitro* and *in vivo* models, organ-on-chip systems, organoids development and new technologies that could be adapted to regenerative ophthalmology and cell therapy applications.

Additionally, we expect to get a cross-disciplinary knowledge exchange by the interaction with experts from diverse biomedical and technological fields, including immunology, tissue engineering, bioinformatics, nanomedicine, advanced imaging, pharmacology and translational medicine.

We also expect to strengthen our translational research pipelines and gain further expertise in clinical trial design, patient-oriented validation strategies, and the clinical implementation of regenerative and advanced therapeutic approaches.

In addition, we expect to have more opportunities to incorporate computational tools, bioinformatics, machine learning and advanced data analysis approaches into experimental and clinical research workflows. And finally, it will help to generate high-impact joint publications, collaborative grant applications and shared training activities that will contribute to the long-term internationalisation and competitiveness of the research group.
Overall, participation would significantly reinforce the group's capacity to develop innovative regenerative medicine approaches while broadening its translational, technological and interdisciplinary reach beyond ophthalmology alone.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Margarita Calonge	Full Professor in Ophthalmology	Department of Surgery, Ophthalmology, ENT, and Physiotherapy, Medical School, UVa IOBA, UVa	Researcher	Experienced	mcalonge@uva.es
Ana de la Mata	Assistant Professor in Immunology	Department of Pediatrics, Immunology, Obstetrics and Gynecology, Nutrition and Food Science, Psychiatry and History of Science, Faculty of Medicine, UVa IOBA, UVa	Researcher	Experienced	ana.mata@uva.es
Teresa Nieto-Miguel	Associate Professor in Cell Biology	Department of Cell Biology, Genetics, Histology and Pharmacology, Faculty of Medicine, Uva IOBA, UVa	Researcher	Experienced	teresa.nieto@uva.es

Main disciplines represented in your team	e.g. ophthalmology, optometry, cell biology, immunology, genetics, biomaterials, data science, engineering, AI, regulatory affairs, marketing and commercialisation. Ophthalmology, cell biology, immunology, regenerative medicine, cell therapy, tissue engineering, stem cells.
Experience in EU / international / multi-partner collaborative projects	Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	☒ Yes ☐ No ☐ Not applicable | Short note: Extensive experience in translational ophthalmology, including the development and clinical application of MSC-based therapies for ocular surface diseases. The IOBA-UVa Clinical Cell Therapy Unit has performed 97 stem cell transplants in patients with limbal stem cell deficiency, including clinical trials and compassionate-use treatments authorised by the Spanish Agency of Medicines and Medical Devices (AEMPS). The group published the first clinical trial showing efficacy of MSC for the human eye. The group also has expertise in minimally invasive

	subconjunctival cell administration and collaborates closely with industrial, regulatory and clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

- Calonge M, Herreras JM, Pérez I, Fernández I, Nieto-Miguel T, López-Paniagua M, Galindo S, Alberca M, García-Sancho J, Sánchez A. *A proof-of-concept clinical trial using mesenchymal stem cells for the treatment of corneal epithelial stem cell deficiency*. Translational Research. 2019. 206: 18-40. https://doi.org/10.1016/j.trsl.2018.11.003

- Pérez I, Galindo S, López-Miguel A, Nieto-Miguel T, de la Mata A, López-Paniagua M, Alberca M, Herreras JM, Calonge M. *In vivo confocal microscopy in limbal stem cell deficiency after mesenchymal stem cell transplantation: A sub-analysis from a phase I-II clinical trial.* Ophthalmology and Therapy. 2023;12(6):3251-3262. https://doi.org/10.1007/s40123-023-00809-7

- Galindo S, López-Paniagua M, de la Mata A, Herreras JM, García-Vázquez B, Marceñido B, Rey E, Higuera-Barón C, Calonge M, Nieto-Miguel T. *Bone marrow- versus adipose tissue-derived mesenchymal stem cells for corneal failure in an experimental model of limbal stem cell deficiency.* Experimental Eye Research 2026;262:110737. https://doi.org/10.1016/j.exer.2025.110737

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***

- Allogeneic mesenchymal stem cells (MSCs) by ocular injection as therapy for graft-versus-host disease refractory to conventional treatments
Funding agency: Instituto de Salud Carlos III (ISCIII) – Independent Clinical Research Projects, Strategic Action in Health 2023
Reference number: ICI23/00027
Dates: 01/01/2024 – 31/12/2027
Role: Partner institution / Local Principal Investigator (Margarita Calonge, IOBA-UVa)
Coordinator: Instituto de Investigación Biomédica de Salamanca (IBSAL). PI: Fermín Sánchez Guijo
Participants: IBSAL, University Hospital of Salamanca, IOBA-University of Valladolid, Gregorio Marañón University Hospital
International: No

- Genetically modified mesenchymal stem cells to improve their potential as ocular surface cell therapy (GenMSCeye)
Funding agency: Spanish Ministry of Science and Innovation – Knowledge Generation Projects (2021 Call)
Reference number: PID2021-125142OA-I00
Dates: 01/09/2022 – 31/08/2026
Role: Beneficiary / Principal Investigators (Ana de la Mata and Alberto López-Miguel)
Funding: €157,300
International: No

- Nano-engineered mesenchymal stem cells to improve the efficacy of stem cell transplants to the ocular surface
Funding agency: Spanish Ministry of Science and Innovation / European Regional Development Fund. Knowledge Generation Projects Oriented to Social Challenges (2019 Call)
Reference number: PID2019-105525RB-100
Dates: 01/06/2020 - 30/11/2025
Role: Beneficiary / Principal Investigators (Margarita Calonge and Teresa Nieto-Miguel)
Participants: Institute of Applied Ophthalmobiology (IOBA), University of Valladolid; CIBER-BBN
International: No

- Aniridia: Networking to address an unmet medical, scientific, and societal challenge
Programme: European Cooperation in Science and Technology (COST Actions), Horizon 2020
Reference number: CA18116
Dates: 11/04/2019 – 10/04/2023
Role: Participant / Collaborating research group (Margarita Calonge, IOBA-UVa)
Coordinator: Neil Lagali (Sweden)

Participants: 27 European and associated countries
International: Yes
European networking action focused on rare ocular diseases, particularly aniridia and limbal stem cell deficiency, promoting collaboration in regenerative medicine, stem cell therapies, tissue engineering and translational ophthalmology.

- ERN-EYE: European Reference Network dedicated to Rare Eye Diseases
Programme: European Reference Network (ERN-EYE), European Commission
Dates: Ongoing (since 2022 participation as affiliated centre)
Role: Affiliated collaborating centre / Clinical and research partner
Coordinator: Prof. Hélène Dollfus (France)
Participants: 52 healthcare providers across 18 EU countries, plus affiliated partners from additional European countries
International: Yes
Participation through the Hospital Clínico Universitario de Valladolid and IOBA-UVa as associated centre. The network promotes cross-border collaboration in diagnosis, clinical management, research and clinical trials for rare eye diseases, including ocular surface disorders and limbal stem cell deficiency.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage. - Clinical Cell Therapy Unit with experience in translational implementation of ATMPs for ocular surface reconstruction. Close collaboration with GMP-compliant production facilities, CROs, industrial partners and regulatory agencies. - Experience in investigator-driven clinical trials and compassionate-use cell-based therapies for ocular surface reconstruction approved by the Spanish Agency of Medicines and Medical Devices (AEMPS). - Expertise in minimally invasive subconjunctival cell administration protocols administration protocols. - Development of clinically relevant rabbit models of limbal stem cell deficiency (LSCD) and ocular surface inflammation, including minimally invasive subconjunctival cell administration protocols. - European patent application related to nanoengineered mesenchymal stromal/stem cells (MSCs) and their therapeutic use as targeted drug-delivery systems for inflammatory and degenerative diseases.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☐ Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:
Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	**Universidad de Valladolid**
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	999862809.
Website(s)	www.ioba.es
Country	SPAIN
Organisation type	x Academic ☐ Non-academic x Hospital / clinic ☐ SME ☐ Large company x Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Edificio IOBA, Campus Miguel Delibes, Paseo de Belén 17 47011 Valladolid, SPAIN
Department(s), institute(s), research group(s) or unit(s) that would be involved	Biomarkers in ocular surface inflammation and ocular pain Group.
Additional institutional resources relevant to collaboration	- Facilities and equipment from IOBA-Uva: https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/ - Centralized services of Uva: https://aplicaciones.educa.jcyl.es/infrared/welcome.action - ICTS Nanbiosis (CIBER-BBN): https://www.nanbiosis.es/

Main contact person for this profile	**María Jesús González García, OD, PhD. University of Valladolid Full Professor**
Main contact email	mariajesus.gonzalez.garcia@uva.es
Other contact person(s), if relevant	Amalia Enríquez-de-Salamanca, PhD. University of Valladolid, Senior Researcher. amalia.enriquez-salamanca@uva.es

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
x Ocular surface	☐ Retina
☐ **Inherited retinal dystrophies (IRDs)**	☐ Age-related ocular disease
☐ **Neuro-vision / integrated vision / neuro-ophthalmology**	☐ Optometry / visual behaviour / rehabilitation
☐ **Advanced therapies / regenerative medicine**	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ **Drug discovery / screening / repurposing**	☐ Ocular vascular disease
☐ **Infection / trauma-related ocular disease**	x Ocular pain / quality of life / patient-reported outcomes
x Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
x Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database **(please provide up to 10)**	**Ocular Surface Inflammation; Dry Eye; Ocular Pain; Tear Biomarkers; Contact Lens; Conjunctiva; Gene and miRNA Expression;**
Geographies of interest for exchanges **(optional)**	

Overview of your research / innovation focus

Our research line, entitled "Biomarkers in Ocular Surface Inflammation and Ocular Pain," is dedicated to the comprehensive phenotyping of patients presenting ocular surface (OS) inflammatory disorders and related alterations, alongside the identification and characterization of molecular biomarkers. These include diagnostic, activity-related, therapeutic, predictive, prognostic, and susceptibility markers, primarily assessed in tear fluid and conjunctival epithelial cells. The work focuses on immune-mediated, chronic inflammatory conditions of the ocular surface, such as Dry Eye Disease (DED), ocular pain syndromes, and contact lens discomfort (CLD), which, due to their severity and/or chronicity, may compromise visual function. A central objective of this research is the development of targeted therapeutic strategies to mitigate these conditions.

Our approach is aligned with the principles of Personalized/Precision Medicine, aiming to elucidate the inflammatory and molecular mechanisms underpinning ocular surface disorders. This is achieved through integrated analyses of cytokine and chemokine profiles in tears, together with gene and microRNA expression in conjunctival epithelial cells, combined with detailed clinical phenotyping of patients.

Our group has significantly contributed to the characterization of cytokine and chemokine expression profiles in tear samples from both healthy individuals and patients, particularly those with DED. These findings have reinforced the concept of an inflammatory component even in moderate or evaporative forms of the disease. Furthermore, we have developed predictive models based on tear biomarkers and gene expression signatures in conjunctival epithelial cells, with demonstrated applicability in DED and in severe chronic ocular manifestations of graft-versus-host disease.

Additionally, our research has advanced the understanding of molecular mechanisms underlying ocular pain, both in DED patients and in post-surgical corneal conditions. In the field of therapeutics, we have demonstrated the efficacy of several natural polyphenolic compounds—such as resveratrol, quercetin, and hydroxytyrosol—in experimental models, both in vitro and in vivo, supporting their potential as novel treatment options for DED.
Finally, our group is actively engaged in the development of molecular-based point-of-care diagnostic devices, designed to facilitate the translation of biomarker research into clinical practice and improve patient management through rapid, personalized assessment.

Why your team could be relevant to a translational eye-health and vision consortium

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

Ocular surface clinical studies and translational research capabilities:

- Highly trained clinical research team with extensive expertise in comprehensive ocular surface evaluation, ensuring high reproducibility and standardisation across studies.
- Advanced experience in in vivo confocal microscopy (IVCM) for the analysis of corneal nerves, immune cells and microstructural changes in ocular surface disorders.
- Expertise in non-contact corneal esthesiometry for the assessment of corneal sensitivity and neurosensory function, enabling detailed characterisation of ocular surface pain and neurotrophic conditions.
- Proven capability in tear film evaporimetry, allowing quantitative assessment of tear film stability and environmental impact on dry eye disease and ocular surface disorders.
- Established protocols for biological sample collection and processing, including tear sampling and conjunctival impression cytology, supporting biomarker discovery and translational research approaches.
- Integrated work within the clinical area of IOBA (University of Valladolid), with direct access to well-characterised patient cohorts through specialised units in:
    - Ocular surface disease
    - Contact lens-related conditions
    - Oculofacial chronic pain
- Strong experience in patient-oriented clinical studies, including observational and interventional designs, with a focus on dry eye disease, contact lens discomfort and ocular surface pain.
- Unique access to and operation within a Controlled Environmental Laboratory (CELab), enabling:
    - Precise modulation of environmental conditions (temperature, humidity, airflow, atmospheric pressure)
    - Standardised and reproducible exposure scenarios
    - Investigation of environmental triggers and their impact on ocular surface physiology and symptoms
- High translational potential through the integration of clinical phenotyping, objective measurements and biological sampling, facilitating links between clinical outcomes, neurosensory mechanisms and molecular biomarkers.

Biological sample processing and Analysis.

- Highly trained personnel with extensive experience in handling and processing patient-derived samples, including tear fluid and conjunctival impression cytology specimens, ensuring high reproducibility and quality control.
- Expertise in multiplexed immunoassays (Luminex/xMAP platform) for the simultaneous quantification of cytokines, chemokines, and other inflammatory mediators in tear samples, enabling high-throughput and systems-level biomarker profiling.
- Advanced capabilities in quantitative real-time PCR (RT-PCR) for the analysis of gene expression and microRNA signatures in conjunctival epithelial cells, supporting mechanistic insights and biomarker validation.
- Design and execution of mechanistic studies using human ocular surface cell lines, enabling controlled investigation of inflammatory pathways, drug screening, and validation of therapeutic targets.
- Ability to correlate molecular findings with detailed clinical phenotyping, supporting biomarker qualification and the development of predictive and diagnostic models.
- Implementation of harmonized protocols for sample collection (tear and impression cytology), processing, and analysis, ensuring consistency across studies and facilitating inter-institutional collaboration.
- Established workflows that bridge laboratory analyses with clinical application, including the identification of biomarker panels suitable for development into point-of-care diagnostic platforms.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- **Ocular Surface Unit and Contact Lens Unit of the IOBA Clinical Area,** where study participants are examined. These units are equipped with all the ophthalmic instruments required for patient evaluation, including:
  slit-lamp biomicroscopes, visual acuity measurement systems, refractive equipment, a corneal topographer, a confocal microscope, and a tear evaporimeter.

- **IOBA Clinical Trials Unit**, which provides support regarding ethical and legal requirements for patient evaluation, as well as data and sample storage.

- **Controlled Environment Chamber (CELab)**, which allows precise control of temperature and humidity in the absence of airflow, ensuring that all participants are assessed under identical environmental conditions. This guarantees standardized data collection and prevents variations attributable solely to fluctuations in environmental parameters.

- **IOBA Molecular Biology Laboratory**, which is equipped with:
  - MAGPIX® multianalyte analysis system and X-Ponent® 3.0 software based on X-MAP technology (Luminex Corporation), as well as the BeadView (Upstate-Millipore) and Belysa (Millipore, Merck) analysis software for multiplex cytokine assays.
  - SpectraMax spectrophotometer and SoftMax® software for ELISA-based determinations, as well as all laboratory materials required for processing and measuring tear samples.
  - Chemidoc MP (BioRad) equipment and ImageLab (BioRad) software for Western Blot protein analysis
  - QuantiStudio 5 (Applied Biosystems) real time-PCR thermoclycler, for gene and miRNA expression analysis.
  - Fluorescence microscope Leica (Leica DMI 6000 B, Leica).

- **IOBA Cell culture Laboratory**, equipped with all necessary equipment for cell culture, including vertical flow cabins and incubators.

- **Instrumental Techniques Laboratory (LTI-INFRARED)**: for confocal microscopy and flow cytometry studies (LTI-INFRARED Platform, https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

- **Structured training environment** for early-stage and postdoctoral researchers, embedded within a multidisciplinary clinical and research setting at IOBA (University of Valladolid), ensuring exposure to both basic science and patient-oriented research.
- **Training in translational ocular surface research**, enabling researchers to integrate clinical phenotyping, advanced diagnostic techniques and molecular biomarker analysis within a precision medicine framework.
- **Hands-on training in advanced clinical and diagnostic techniques**, including:
  - In vivo confocal microscopy (IVCM) for corneal nerve and immune cell analysis
  - Non-contact corneal esthesiometry for neurosensory function assessment
  - Tear film evaporimetry and ocular surface functional testing
- **Advanced training in biological sample collection and processing**, including tear sampling and conjunctival impression cytology, linked to downstream molecular analyses (protein, gene and miRNA expression), including:
  - Hands-on expertise in protein extraction, quantification, and characterization using techniques such as Western blotting, spectrophotometric methods, immunofluorescence, ELISA, and multiplexed Luminex xMAP® technology for high-throughput, multi-analyte profiling.
  - Gene and miRNA expression analysis: Comprehensive training in nucleic acid workflows, including mRNA and miRNA extraction, quantification, and quantitative real-time RT-PCR, enabling precise characterization of molecular signatures associated with ocular surface diseases.
- **Exposure to controlled-environment experimental research** through the CELab (Controlled Environmental Laboratory), allowing incoming staff to design and perform studies under strictly regulated environmental conditions.
- **Strong capacity to train researchers in patient-based clinical studies**, including study design, protocol implementation, ethical considerations and interaction with well-characterised cohorts from specialised units (ocular surface disease, contact lens, and oculofacial chronic pain).
- **Training in translational research approaches**, bridging clinical data with molecular biomarkers and supporting the development of diagnostic tools and therapeutic strategies.

- **Opportunities for interdisciplinary knowledge exchange**, including interaction with clinicians, biologists, optometrists and data analysts.
- **Ocular surface cell culture models**: Training in the maintenance and experimental use of human ocular surface cell lines, including the design of in vitro assays to investigate inflammatory mechanisms and evaluate therapeutic compounds.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. Again, please think broadly and beyond ophthalmology.

- Access to complementary cutting-edge methodologies and technologies not currently available within our institution, particularly in areas such as proteomics, advanced multi-omics integration, AI-assisted data analysis and digital health tools applied to ocular surface disease.
- Strengthening of translational research pathways, enabling a more efficient progression from biomarker discovery to clinical validation and eventual implementation in diagnostic and therapeutic strategies.
- Opportunities to collaborate with non-academic partners (SMEs, industry, and technology developers), particularly in the areas of point-of-care diagnostic devices, biomarker-based tools and innovation-oriented product development.
- Expansion of our expertise in patient-centred research approaches, including the development and validation of Patient-Reported Outcome Measures (PROMs) and their integration with objective clinical and molecular data.
- Access to diverse patient populations and clinical settings, enhancing the robustness and external validation of our research findings.
- Enhancement of our capabilities in data science, bioinformatics and advanced statistical modelling, particularly for the integration of multidimensional datasets (clinical, molecular and environmental).
- Exposure to different regulatory, ethical and clinical frameworks across countries, improving our capacity to design and implement multicentre international clinical studies.
- Strengthening long-term strategic collaborations and international networks, facilitating joint publications, future EU proposals and sustainable knowledge exchange beyond the duration of the project.
- Training and career development opportunities for our researchers through intersectoral and interdisciplinary mobility, particularly benefiting early-career and postdoctoral researchers.
- Increased capacity to co-develop and validate environment-sensitive research paradigms, building on our Controlled Environmental Laboratory (CELab) expertise through collaboration with partners working on environmental health, exposure science and related fields.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Maria Jesús González García	Full Professor. Principal Investigator	IOBA-UVa	Researcher	Experienced	mariajesus.gonzalez.garcia@uva.es
Amalia Enríquez de Salamanca Aladro	Senior Scientist. Principal Investigator	IOBA-UVa	Researcher	Experienced	amalia.enriquez-salamanca@uva.es
Alberto López Miguel	Full Professor, Principal Investigator	IOBA-UVa	Researcher	Experienced	alberto.lopez.miguel@uva.es
Itziar Fernández Martínez	Associate Professor / Principal Investigator	IOBA-UVa	Researcher	Experienced	itziar.fernandez@uva.es
Marta Blanco Vázquez		IOBA-UVa	Researcher	Middle-Stage	marta.blanco.vazquez@uva.es
Alberto López de la Rosa	Associate Professor	IOBA-UVa	Researcher	Experienced	alberto.lopez.rosa@uva.es
Andrés Ángel Calderón García	Assistant Professor	IOBA-UVa	Researcher	Middle-Stage	andresangel.calderon.garcia@uva.es

Elena Martínez Plaza	Postdoctoral researcher	IOBA-UVa	Researcher	Middle-Stage	elena.martinez.plaza@uva.es
Cristina Arroyo del Arroyo	Postdoctoral researcher	IOBA-UVa	Researcher	Middle-Stage	cristina.arroyo.arroyo@uva.es
Cristina Valencia Sandónis	Predoctoral Student	IOBA-UVa	Researcher	Early Stage	cristina.valencia@uva.es
Ivonne Guachamin	Predoctoral Student	IOBA-UVa	Researcher	Early Stage	ivonne.guachamin25@estudiantes.uva.es
Lidia Cocho Archiles	Ophtalmologist	HCU/ IOBA-UVa	Opthalmologists	Experienced	lidiacocho@hotmail.com

Main disciplines represented in your team	**e.g. ophthalmology, optometry, cell biology, immunology, genetics, biomaterials, data science, engineering, AI, regulatory affairs, marketing and commercialisation.** Ophthalmology, Optometry, Biochemistry, Molecular Biology, Cell biology, Biostatistics/Data science, Clinical trials,
Experience in EU / international / multi-partner collaborative projects	• Implementation of molecular biomarker-based biosensor for patient stratification and tailored therapy in inflammatory-dry eye disease. Stratif-EYE. Proposal number 754620-1. Entidad financiadora: Programa H2020. Call H2020-SC1-2017. (not granted). Coordinators: CIBER BBN IOBA-UVa CB06/01/0003 Group: Leader: A. Enríquez-de-Salamanca (IOBA). Partners: 1) Centro De investigación biomédica En Red (España) (Coordinadores); 2) Tampereen Yliopisto (Finlandia); 3) Klinikum Der Universitaet Zu Koeln (Alemania); 4) Vision I Mas D Sl (España); 5) Asper Biotech As (Estonia); 6) Asphalion Sl (España); 7) Innova Srl (Italia); 8) Asociación Española Síndrome De Sjögren (Aess) (España); 9) Bundesverband-Auge E.V (Alemania • IT-DED 3 MSC EU project. María Jesus González-García UVa Pirncipal investigator; WP4 Leader,ERS6 Co-Supervisor. Amalia Enríquez-de-Salamanca WP1 Leader; ESR4 Co supervisor; Supervisory board member. • CHOPIN MSC Doctoral Network EU project 2024, 2025 and 2026 calls application: Not Granted • COST ACTION 2025. Granted (ACTION CA 24112 "Tear precision", 2025-2029).(www.tear-precision.eu) • Tear Research Network (TRN; www. Tearresearchnetwork.com)
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

- Vázquez A, Martínez-Plaza E, Fernández I, Sobas EM, González-García MJ, Enríquez-de-Salamanca A, Ortega E, López-Miguel A, Calonge M. Phenotypic characterization of patients developing chronic dry eye and pain after refractive surgery: A cross-sectional study. Ocul Surf. 2022 Oct;26:63-74. doi: 10.1016/j.jtos.2022.07.010.

- Calderón-García AÁ, Valencia-Nieto L, Valencia-Sandonis C, López-de la Rosa A, Blanco-Vazquez M, Fernández I, García-Vázquez C, Arroyo-Del Arroyo C, González-García MJ, Enríquez-de-Salamanca A . Gene expression changes in conjunctival cells associated with contact lens wear and discomfort. Ocul Surf. 2024 Jan;31:31-42. doi: 10.1016/j.jtos.2023.12.004

- Calderón-García AÁ, López-de la Rosa A, Valencia-Nieto L, García-Vázquez C, Arroyo-Del-Arroyo C, López-Miguel A, Blanco-Vázquez M, Novo-Diez A, Fernández I, Enríquez de Salamanca A, González-García MJ . Tear Cytokines as Predictive Biomarkers of Success in Contact Lens Discomfort Management. Ocul Immunol Inflamm. 2024 Dec;32(10):2339-2347. doi: 10.1080/09273948.2024.2361114

- Blanco-Vázquez, M.;Novo-Diez, A.; Vázquez, A.; Enríquez-de-Salamanca, A.;González-García, M.J.; Calonge, M. Clinical Characterization of the Lacrimal Functional Unit in Patients with Chronic Ocular Pain Associated with Dry Eye Disease. J. Clin. Med.2025, 14, 5250. https://doi.org/10.3390/jcm14155250.

- Blanco-Vázquez M, Vázquez A, Fernández I, Novo-Diez A, Martínez-Plaza E, García-Vázquez C, González-García MJ, Sobas EM, Calonge M, Enríquez-de-Salamanca A . Inflammation-related molecules in tears of patients with chronic ocular pain and dry eye disease. Exp Eye Res. 2022 Jun;219:109057. doi: 10.1016/j.exer.2022.109057.

- Cocho L, Fernández I, Calonge M, Sainz de la Maza M, Rovira M, Stern ME, Garcia-Vazquez C, Enríquez-de-Salamanca A. Prehematopoietic Stem Cell Transplantation Tear Cytokines as Potential Susceptibility Biomarkers for Ocular Chronic Graft-Versus-Host Disease. Invest Ophthalmol Vis Sci. 2017 Sep 1;58(11):4836-4846. doi: 10.1167/iovs.17-21670.

Up to 5 relevant previous projects, grants, networks or collaborations

1) **IT-DED 3 MSC EU project (INTERNATIONAL)**
Integrated Training in Dry Eye Disease Drug Development (IT-DED3)
Grant agreement number 765608
Entidad financiadora: EUROPA PROGRAMA H2020
Call: H2020-MSCA-ITN-2017; Funding scheme: MSCA-ITN-ETN
2018-2022
- Coordinators: UNIVERSITEIT ANTWERPEN (BELGICA).
- Beneficiaries: UNIVERSITEIT ANTWERPEN (BELGICA); Universite Pierre Et Marie Curie - Paris 6 (Francia); UNIVERSIDAD DE VALLADOLID (ESPAÑA) : Maria J González García: UVA IP,WP4 Training leader; ESR3 co-supervisor. Amalia Enríquez de Salamanca: WP1 Drug Discovery leader; Supervisory board member; ESR4 Co-supervisor

Latvijas Organiskas Sintezes Instituts (Letonia); Ita-Suomen Yliopisto (Finlandia); Klinikum Der Universitaet Zu Koeln (Alemania); Instituto De Biologia Experimental E Tecnologica (Portugal)

- Partners: Quintessence Consulting NV; Mercachem BV; HG Beyond; Riga Technical University; New University of Lisbon; Novaliq GmbH DE; Centre Hospitalier National des Quinze-Vingts; Horus Pharma; Heildelberg Enginieering.

Total amount granted: 3,002,689.44€ (UVA : 495,745.92€)

2) **COST Action CA24211 (INTERNATIONAL):**
Implementation of Tear Fluid Biomarkers in Precision Medicine. TEAR-Precision (COST ACTION; Open Call Collection OC-2024-1
Proposal Reference OC-2024-1-27225
REF : Action N. CA24112
COST European Cooperation in Science and technology. COST Association AISBL. European Union.
2025-2029.4 años
IP: Marlies Gijs. Maastricht University.
Main Proposer: The Netherlands
Network of Proposers:
-Full Member: Belgium, Croatia, Czech Republic, Denmark, Estonia, France, Germany, Greece, Hungary,Italy, Lithuania, Moldova, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Spain, Sweden,Switzerland, Türkiye, United Kingdom.
- Cooperating Member: Israel
- International Partner: Australia, Canada, Colombia, Hong Kong SAR, India, New Zealand, Singapore,United States
- Specific Organisations :EU Institutions, Bodies, Offices and Agencies (EC/EU): University of Crete Medical School
- International Organisation: Association for Research in Vision and Ophthalmology; Schepens Eye Research Institute; Schepens Eye Research Institute; Tan Tock Seng Hospital; Vision Research Foundation, Sankara Nethralaya; Cincinnati Children's Hospital Medical Center
-Industrial Dimension: SMEs: Canada, Netherlands, Switzerland, United States. Large companies: Switzerland
Spain: Amalia Enríquez de Salamanca (FUNGE), MC member for Spain; (WG 4 co-lider); María J González-García (UVA)

Amount granted: 600000€

3) **CIBER-BBN (Bioengineering, Biomaterials and Nanomedicine Research Network)(National)**
IOBA-UVa CB06/01/0003 Group. Instituto de Salud Carlos III, Ministerio de Sanidad y Consumo.
Group Leader Margarita Calonge Cano.
01/2007-actualidad
https://www.ciber-bbn.es
The CIBER in its thematic area of Bioengineering, Biomaterials and Nanomedicine (CIBER-BBN) is currently composed of 45 research groups (43 full members, 1 associated group, and 1 linked clinical group), selected on the basis of their scientific excellence. These groups work primarily within three scientific programs: 1) Bioengineering and Medical Imaging, 2) Biomaterials and Advanced Therapies, and 3) Nanomedicine. The center's activities are aimed both at developing systems for prevention, diagnosis, and monitoring, and at advancing technologies related to specific therapies, such as regenerative medicine and nanotherapies. CIBER-BBN has been carrying out its work since 2007, providing support, training, infrastructures, and resources to researchers, thereby contributing to the promotion of scientific research in our country. The overarching goals outlined in the CIBER-BBN Strategic Plan for the 2018–2021 period include maintaining the scientific and technological excellence of the thematic area, fostering collaborative research among CIBER-BBN groups, promoting translational research, and facilitating the transfer of research results to the productive sector.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

1. **Title: COMPOSICIÓN PARA SU USO EN EL TRATAMIENTO Y/O PREVENCIÓN DE LA INFLAMACIÓN, EL ESTRÉS OXIDATIVO Y LA NEOVASCULARIZACIÓN OCULAR**
   REF: ES 2 428 665 B1
   Inventores: González García, MJ; Calonge Cano M, Enríquez-de-Salamanca Aladro, A; Stern, ME; Avengózar Vela, A.
   N. de solicitud: P201230664 País de prioridad: España Fecha de prioridad: 04-Mayo 2012
   Entidad titular: Universidad de Valladolid, Valladolid España; Allergan Inc,California, EEUU

2. **Registro de la propiedad intelectual del texto científico "Contact Lens Discomfort Index" ,** Universidad de Valladolid.
   Autores (p.o. de firma): Arroyo del Arroyo, C; Fernández Martínez, I; López-Miguel A; González García, MJ, López de la Rosa, A; Enríquez-de-Salamanca Aladro, A
   01/03/2019
   N. de solicitud: VA-71-2019 País de prioridad: España

3. **Title: COMPUESTOS PARA SU USO EN LA PREVENCIÓN Y/O TRATAMIENTO DE LA INFLAMACIÓN Y EL ESTRÉS OXIDATIVO DEL SEGMENTO ANTERIOR DEL OJO.**
   Inventores (p.o. de firma): Margarita Calonge Cano, Maria José Cocero, María Jesús González García, SORAYA Rodriguez Rojo , Nikolaos Katsinas, Enríquez-de-Salamanca Aladro, A;;.
   REF : ES2080.29
   N. de solicitud P202031306 País de prioridad: España Fecha de depósito: 28 diciembre 2020
   Entidad titular: Universidad de Valladolid, Valladolid España.

4. **Clinical Protocol: "Enfermedad de ojo seco: detección y manejo",** González García, MJ, Blanco Vázquez M, Arroyo del Arroyo C, Calonge Cano M.. Editorial: Grupo ICM de comunicación, Madrid 2026. ISBN: 979-13-990617-8-9.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	**x Yes ☐ Maybe ☐ No capacity to send staff on secondment**
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles

Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

- **Strong translational and multidisciplinary capacity in ocular surface diseases**: Integration of clinical expertise (ophthalmology and optometry) with molecular biology, biochemistry, and biostatistics, with a specific focus on Dry Eye Disease, contact lens discomfort, and ocular pain, enabling robust patient phenotyping and direct linkage to underlying molecular mechanisms.
- **Advanced expertise in biomarker discovery and molecular analysis**: Proven capabilities in tear and conjunctival sample processing and analysis using multiplex xMAP (Luminex), ELISA, and RT-PCR for gene and miRNA profiling, supporting the identification and validation of clinically relevant biomarkers in dry eye, contact lens-related conditions, and ocular pain.
- **Established experimental and innovation platform with translational impact**: Extensive experience in in vitro ocular surface models and therapeutic evaluation, particularly in Dry Eye Disease, combined with the development of molecular-based point-of-care diagnostic tools to facilitate clinical implementation and personalized management.

Capabilities or expertise you would especially like to access through the consortium

Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium.

We would like to connect with other research teams or clinical units with expertise in or related to;

- Proteomic analysis facilities.
- Pain units
- Ocular Surface research/clinical groups.
- Point of Care development companies
- Patient-related output measurements (PROMs) development
- Data mining/analysis

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	**x Yes** ☐ **No**

x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: María Jesús González García and Amalia Enríquez de Salamanca Aladro
Date completed: 29th June, 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Valladolid
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	
Website(s)	https://www.uva.es/export/sites/uva/
Country	SPAIN
Organisation type	X Academic ☐ Non-academic X Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	
Department(s), institute(s), research group(s) or unit(s) that would be involved	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid
Additional institutional resources relevant to collaboration	

Main contact person for this profile	Associate Prof. Dr. SALVADOR PASTOR IDOATE and Prof. RICARDO USATEGUI MARTÍN

Main contact email	Salvador.pastor@uva.es and ricardo.usategui@uva.es
Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
X Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
X Clinical trials / translational validation	☐ Other (please specify):

Free-Keywords for internal searchable database (please provide up to 10)	retinal organoids; neuroinflammation; AI diagnostics; drug delivery; biomarkers; neuroprotection; retina; retinal detachment, PVR, neurodegeneration; Gene therapy; cell therapy
Geographies of interest for exchanges (optional)	EU/Associated Countries; UK sociated Countries; UK; North America; Latin America; Asia

Overview of your research / innovation focus
Our team focuses on translational retinal research, with a particular emphasis on rhegmatogenous retinal detachment, proliferative vitreoretinopathy, neuroinflammation, photoreceptor degeneration and precision medicine in vitreoretinal surgery. Over the last years, our group has developed a clinically oriented research line linking genetic susceptibility, inflammatory pathways, retinal imaging biomarkers and functional outcomes after retinal detachment surgery. A central element of this work is the RAISE-VISION programme, which aims to move from conventional prognostic assessment towards personalised risk stratification based on molecular biomarkers, multimodal imaging and artificial intelligence. Our recent work has identified and validated the TP53 Arg72Pro polymorphism as a prognostic biomarker associated with visual and neurodegenerative outcomes after retinal detachment, supported by both human clinical data and humanised mouse models. The next translational step is to integrate rapid point-of-care genotyping, clinical and OCT/OCTA imaging variables into explainable AI-based decision-support tools. The ultimate goal is to identify high-risk patients early, support surgical prioritisation, guide intensified follow-up and enable genotype-informed neuroprotective strategies. The team combines clinical retina expertise, molecular and cellular biology, preclinical models, retinal organoids, imaging analysis, AI development and medical-device translation.

Why your team could be relevant to a translational eye-health and vision consortium
Our team could contribute strong added value to a translational eye-health and vision consortium because it bridges clinical ophthalmology, molecular biomarker discovery, preclinical validation and enabling technologies for real-world implementation. The group is embedded in a hospital–academic environment involving the Hospital Clínico Universitario de Valladolid, IOBA–University of Valladolid, IBioVALL and the RICORS-REI network, providing access to well-characterised retinal detachment cohorts, multimodal imaging, surgical expertise and translational laboratory platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

• Clinical and translational expertise in vitreoretinal diseases, particularly rhegmatogenous retinal detachment, proliferative vitreoretinopathy, photoreceptor degeneration and functional recovery after retinal surgery.
• Longstanding experience in genetic risk stratification for PVR and poor visual outcomes, including TP53 Arg72Pro, TNF, TNFR2, SMAD7, PIK3, LTA, BAX and BCL-2-related pathways.
• Prospective clinical phenotyping of retinal detachment patients, including surgical variables, macular status, timing from symptoms, anatomical outcomes, PVR development and longitudinal functional recovery.
• Multimodal retinal imaging analysis, including OCT, OCT-A, autofluorescence, wide-field imaging and structural biomarkers such as photoreceptor integrity, ellipsoid zone/ELM status, LPOS, hyperreflective dots and cystoid macular changes.
• Molecular and cellular profiling of retinal detachment, including genotyping, transcriptomic analysis, cytokine/inflammatory profiling, apoptosis/autophagy markers and immune-cell characterisation.
• Preclinical validation using humanised TP53 Arg72Pro knock-in mouse models and retinal organoid/ex vivo systems to explore genotype-dependent neurodegeneration and neuroprotective strategies.
• Development of AI-based predictive models integrating clinical, imaging and genetic data for personalised risk stratification and clinical decision support.
• Translational development of point-of-care diagnostic tools, including Lab-on-a-Chip / rapid genotyping approaches designed for perioperative or early clinical use.
• Experience in clinical implementation, regulatory interaction, ethics submissions, MDR/IVDR-oriented device translation and health-technology adoption pathways.
• Capacity to connect clinical retina specialists, molecular scientists, engineers, AI developers, biotech partners and public healthcare networks within a single translational workflow.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Hospital-based vitreoretinal surgery setting at the Hospital Clínico Universitario de Valladolid, enabling prospective recruitment, clinical phenotyping and real-world validation of diagnostic or decision-support tools.

IOBA–University of Valladolid and IBioVALL translational research environment, combining clinical ophthalmology, molecular biology, imaging analysis and biomedical innovation.

Existing well-characterised retinal detachment cohort with clinical, imaging and genetic data, including the RICORS-Retina4 dataset used for AI model development and validation.

Access to multimodal ophthalmic imaging platforms, including OCT/OCT-A, fundus imaging, wide-field retinal imaging and image-based biomarker extraction.

Molecular and cellular biology laboratories for genotyping, inflammatory biomarker analysis, cytokine profiling, apoptosis/autophagy studies and validation of candidate biomarkers.

Biobank-supported management of biological samples, including blood, vitreous and retinal material obtained under approved ethical protocols.

Preclinical platforms including humanised TP53 Arg72Pro mouse models and retinal organoid/ex vivo models for mechanistic and therapeutic validation.

AI and data-science collaboration capacity through engineering and computer-science partners, including development of explainable machine-learning models and clinical dashboards.

Point-of-care diagnostic development capacity through collaboration with biotech and engineering partners for Lab-on-a-Chip, LAMP-based or immunochromatographic systems.

Regulatory, ethics and translational support through CEIm-approved clinical research, IBioVALL/UVa structures, OTRI support and previous experience in medical-device and software-as-medical-device translation.

Training capacity and knowledge-transfer potential

Inbound researchers, clinicians, engineers or innovation fellows could receive practical training in a highly translational ophthalmology environment, moving from patient care to biomarker discovery, preclinical validation and clinical implementation. Specifically, they could learn:
• Clinical pathways in retinal detachment and proliferative vitreoretinopathy, including surgical decision-making, timing of intervention and follow-up strategies.
• Standardised collection of clinical, imaging and biological data in prospective retinal cohorts.
• Interpretation of multimodal retinal imaging and extraction of clinically meaningful OCT/OCT-A biomarkers.
• Genetic and molecular biomarker workflows, including sample handling, genotyping strategies, inflammatory profiling and integration with clinical outcomes.
• Design and validation of AI-based prediction models for ophthalmology, including explainability, clinical usability and integration into decision-support workflows.
• Translational use of humanised animal models and retinal organoid/ex vivo systems to test genotype-dependent mechanisms and neuroprotective interventions.
• Development and early validation of point-of-care diagnostic devices for ophthalmic use.
• Practical aspects of ethics approval, GDPR-compliant data management, biobanking, MDR/IVDR-oriented device development and preparation for clinical translation.
• Knowledge transfer in project coordination, multicentre collaboration, public healthcare implementation, dissemination, guideline input and health-economic evaluation.

What your team would hope to gain from participating

Through participation in a translational eye-health and vision consortium, our team would aim to strengthen the international validation, scalability and clinical adoption of our precision-medicine approach in retinal detachment. We would particularly value:

- Access to complementary patient cohorts for external validation of genetic, imaging and AI-based risk prediction models.
- Collaboration with centres experienced in advanced retinal imaging, automated image segmentation, digital biomarkers and clinical AI deployment.
- Access to additional omics, single-cell, proteomic or spatial-biology platforms to deepen mechanistic understanding of neuroinflammation, apoptosis, fibrosis and photoreceptor degeneration.
- Collaboration with groups working on retinal organoids, advanced preclinical models, drug screening and neuroprotective or anti-inflammatory therapeutic strategies.
- Industrial and engineering expertise to accelerate the maturation of point-of-care genotyping devices, microfluidics, diagnostic assays and software-as-medical-device development.
- Support in regulatory strategy, IVDR/MDR compliance, quality-management systems, usability testing and health-technology assessment.
- Opportunities for joint publications, shared datasets, harmonised clinical protocols and multicentre grant applications.
- Exposure to broader translational pathways across Europe, including patient-oriented validation, implementation science, payer perspectives and public healthcare adoption.
- Long-term strategic collaboration to transform RAISE-VISION from a strong local/national translational programme into a scalable European precision-medicine platform for retinal disease.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
SALVADOR PASTOR IDOATE	Associate Prof. Dr.	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid			Salvador.pastor@uva.es
RICARDO USATEGUI MARTÍN	Prof.	Group name: Biomarkers in Neurodegenerative, Metabolic, and Inflammatory Pathology Department of Cell Biology. Faculty of Medicine. University of Valladolid IOBA- University of Valladolid			ricardo.usategui@uva.es

Main disciplines represented in your team	Ophthalmology and vitreoretinal surgery; translational retinal research; cell biology; molecular biology; genetics/genomics; immunology and neuroinflammation; retinal imaging and image analysis; preclinical models and retinal organoids; biomaterials and microfluidics; biomedical engineering; artificial intelligence/data science; clinical epidemiology and biostatistics; regulatory affairs, ethics and medical-device translation.
Experience in EU / international / multi-partner collaborative projects	The team has experience in multicentre and international retinal research through the Retina 4 Project, the RICORS-REI network, IBioVALL/IOBA-University of Valladolid collaborations, and formal partnerships with leading European ophthalmology centres in Spain, Italy, Germany and the United Kingdom. The group also participates in ERN-EYE-related activities and has coordinated translational projects combining clinical cohorts, genetic biomarkers, preclinical models, AI and point-of-care diagnostic development.
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
• Publications in peer-reviewed journals: 53. 22 in Q1 journals https://pubmed.ncbi.nlm.nih.gov/?term=pastor-idoate+S 66. 26 in Q1 journals. https://pubmed.ncbi.nlm.nih.gov/?term=usategui-martin+R • ORCID iD (SPI): 0000-0002-6463-4227 and (RUM): 0000-0001-7699-4388 • Web of Science Researcher ID (SPI): IAM-1368-2023. H-Index: 14; (RUM):): IAM: 56245265700 H-Index: 18 • Salvador Pastor: Two Research Six-Year periods (2012-2023) recognized by the National agency for Quality Assessment of Spain (ANECA). Ricardo Usategui: Two Research Six-Year periods by ANECA • The 5 most relevant publications: 1- The p53 codon 72 polymorphism (rs1042522) is associated with proliferative vitreoretinopathy: the Retina 4 Project. Ophthalmology. 2013 Mar;120(3):623-628. doi: 10.1016/j.ophtha.2012.08.019. Epub 2012 Dec 1. PMID: 23207172. 2- BAX and BCL-2 polymorphisms, as predictors of proliferative vitreoretinopathy development in patients suffering retinal detachment: the Retina 4 project. Acta Ophthalmol. 2015 Nov;93(7):e541-9. doi: 10.1111/aos.12718. Epub 2015 May 19. PMID: 25991504. 3- The T309G MDM2 gene polymorphism is a novel risk factor for proliferative vitreoretinopathy. PLoS One. 2013 Dec 9;8(12):e82283. doi: 10.1371/journal.pone.0082283. PMID: 24349246; PMCID: PMC3857251. 4- Proliferative vitreoretinopathy: A new concept of disease pathogenesis and practical consequences. Prog Retin Eye Res. 2016 Mar;51:125-55. doi: 10.1016/j.preteyeres.2015.07.005. Epub 2015 Jul 21. PMID: 26209346. 5- Galindo-Cabello N, Sobas-Abad EM, Lapresa R, Agulla J, Almeida Á, López A, Pastor JC, Pastor-Idoate S, Usategui-Martín R. The TP53 Arg72Pro polymorphism predicts visual and neurodegenerative outcomes in retinal detachment. Cell Death Dis. 2025 May 26;16(1):415. doi: 10.1038/s41419-025-07739-1. PMID: 40419469; PMCID: PMC12106684.

Up to 5 relevant previous projects, grants, networks or collaborations
Beneficiaries / participating investigators: Prof. Salvador Pastor-Idoate (SPI) and Prof.. Ricardo Usategui-Martín (RUM) Relevant funded projects and research networks: 1. PID2020-114585RA-I00 – National R&D project funded by the Spanish State Research Agency. 2. PID2023-147958OB-I00 – National R&D project funded by the Spanish State Research Agency. 3. RD21/0002/0017 – RICORS-REI Network – Related research groups: GR10, GR11, GR14, GR32 and GR41. 4. RD24/0001/0008 – RICORS-REI Network – Related research groups: GR10, GR11, GR14, GR32 and GR41. 5. PDC2023-145857-I00 / PDC2023-35 145857-100 – Proof-of-concept / translational innovation project. 6. CLU-2023-1-04 “Escalera de Excelencia” – Excellence programme supporting strategic research capacity and institutional strengthening.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
N/A

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation’s/team’s capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes X Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff X Other profiles: Clinical; Innovation / business; Data / software
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Original rough estimates provided: Sending capacity: 1–2 staff, depending on funding availability and clinical/teaching workload; approximately 1–4 person-months in total, preferably through short stays or split visits; short stays of 2–4 weeks; exceptionally 1–2 months if fully funded and compatible with clinical/academic duties; clinical researchers, clinician-scientists, early-career researchers, data/imaging collaborators, innovation-oriented staff. Hosting capacity: 1–3 incoming staff, depending on profile, supervision needs and available workspace; approximately 1–6 person-months in total, preferably through short stays or split visits; short stays of 2–8 weeks; flexible split stays would be preferred over long continuous stays; clinical researchers, preclinical/translational researchers, technical staff, data/software profiles, innovation/business development profiles. Incoming and outgoing mobility would need to be assessed case by case according to institutional rules of the University of Valladolid, Hospital Clínico Universitario de Valladolid, IOBA and/or IBioVALL, as applicable.

	Our team is open to staff exchange, but mobility should be planned as externally funded, short-term and clearly task-oriented, as the organisation cannot normally provide salary support for incoming or outgoing staff.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Clinically embedded translational retina team with strong expertise in retinal detachment, proliferative vitreoretinopathy, vitreoretinal surgery, visual outcomes and real-world healthcare implementation.

Precision-medicine approach in retinal disease, integrating genetic biomarkers, neuroinflammation, photoreceptor degeneration, multimodal imaging and functional outcomes.

Validated biomarker and cohort assets, including experience with TP53 Arg72Pro and other genetic predictors of PVR/visual prognosis, supported by clinical cohorts, imaging data and preclinical validation.

AI and point-of-care innovation pipeline, focused on combining clinical variables, OCT/OCTA imaging and rapid genetic testing into explainable decision-support tools for early risk stratification.

Strong translational ecosystem, linking hospital care, IOBA–University of Valladolid, IBioVALL, RICORS-REI, preclinical models, organoid research, engineering/data-science partners and medical-device-oriented development.

Capabilities or expertise you would especially like to access through the consortium

We would particularly like to connect with partners offering complementary expertise in:

• External clinical validation cohorts in retinal detachment, proliferative vitreoretinopathy, inherited retinal disease or other neuroinflammatory retinal conditions.

• Advanced retinal imaging and digital biomarkers, including automated OCT/OCTA segmentation, image harmonisation, quantitative imaging pipelines and multimodal imaging repositories.

• AI, explainable machine learning and clinical decision-support deployment, especially teams experienced in trustworthy AI, federated learning, data interoperability, clinical dashboards and software-as-medical-device pathways.

• Point-of-care diagnostics and microfluidics, including Lab-on-a-Chip development, LAMP/qPCR miniaturisation, biosensors, assay validation, usability testing and IVDR-oriented diagnostic translation.

• Omics and spatial biology platforms, including single-cell transcriptomics, proteomics, metabolomics, spatial transcriptomics and immune-cell profiling in ocular tissues.

• Preclinical and organoid-based therapeutic testing, particularly teams working on retinal organoids, ex vivo retinal models, neuroprotection, anti-inflammatory therapies, anti-fibrotic strategies and drug screening.

• Regulatory, health-economic and implementation-science expertise, including MDR/IVDR strategy, quality management systems, health technology assessment, payer perspectives and adoption within public healthcare systems.

• Industry and innovation partners interested in co-developing scalable diagnostic, digital-health or precision-medicine tools for ophthalmology

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Overall, our team is strongly interested in participating in a translational eye-health and vision consortium, particularly where activities are clearly task-oriented, adequately funded, ethically approved and aligned with our clinical, scientific and innovation priorities in precision retinal medicine.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC)	999862809
Website(s)	www.ioba.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Paseo de Belén 17, 47011, Valladolid, España
Department(s), institute(s), research group(s) or unit(s) that would be involved	Nanotechnological Solutions for Ocular Functional Recovery (NANO-SOS-4EYE)
Additional institutional resources relevant to collaboration	

Main contact person for this profile	Yolanda Diebold, PhD, Full Professor Ophthalmology, University of Valladolid
Main contact email	yolanda.diebold@uva.es
Other contact person(s), if relevant	Laura García-Posadas, PhD, laura.gacia.posadas@uva.es

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	conjunctiva; cornea; drug delivery; extracellular vesicles; ocular surface inflammation; oxidative stress; regeneration; secretome; tissue engineering
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
Our focus is therapeutic innovation to treat ocular surface inflammatory diseases. In one hand, we use nanotechnology to design novel anti-inflammatory and immunomodulatory nanomedicines. On the other hand, we develop more complex and compliant with the 3Rs in vitro and ex vivo models in which our therapeutical developments can be reliably tested. Currently, our work is focused on the development of 3D models of human cornea and conjunctiva, the study of the therapeutic potential of cornea and conjunctiva-derived extracellular vesicles (EVs), and the design of novel drug delivery platforms combining new generation nanoparticles and EVs.

Why your team could be relevant to a translational eye-health and vision consortium
Our team would bring to such a consortium our expertise in testing the potential of active molecules and/or novel nanomedicines in relevant human-derived 2D and 3D in vitro models, and in ex vivo porcine and rabbit eye models. Also, our know-how about ocular surface tissue-derived EVs in terms of isolation, enrichment, characterization and functional assays. Finally, we have knowledge and resources to develop nanomedicines using drug delivery platforms and tissue engineering strategies combining cells and different biomaterials.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
• *Cell culture (2D, 3D, primary cells, immortalized cell lines, mesenchymal stromal cells)* • *ELISA, WB analysis, Multiplex analysis, Q-PCR, immunofluorescent staining, histopathology, Confocal and AFM microscopies, DLS* • *In vitro models for human cornea and conjunctiva (normal, inflamed, under oxidative stress wound healing)* • *Ex vivo ocular surface models from rabbit and pig enucleated eyes*

Infrastructure, equipment, facilities or datasets relevant to consortium work
• *In-house specialized laboratories for cell culture, histopathology, protein and genetic material analyses; access to Laboratorio de Técnicas Instrumentales from the University of Valladolid*

- *Access to ocular surface human tissues*
- *Access to enucleated rabbit and pig eyes from a certified slaughterhouse*

Training capacity and knowledge-transfer potential
• *Scientific, technical, and translational skills* • *Soft skills*

What your team would hope to gain from participating
Access to methods and technologies, patient-oriented validation, joint publications and longer-term strategic collaboration.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Yolanda Diebold	PI	IOBA-UVa / NANO-SOS-4EYE	Full Professor	Senior (R4)	yolanda.diebold@uva.es
Laura García-Posadas	Co-PI	IOBA-UVa / NANO-SOS-4EYE	Associate Professor	Senior (R3)	laura.garcia.posadas@uva.es
Marta Mínguez-Moratinos	Predoctoral student	IOBA-UVa / NANO-SOS-4EYE	Predoctoral student	Early (R1)	
Guillermo López-Salmerón	Predoctoral student	IOBA-UVa / NANO-SOS-4EYE	Predoctoral student	Early (R1)	

Main disciplines represented in your team	Cell biology, immunology, biotechnology, biomaterials, pharmaceutical technology
Experience in EU / international / multi-partner collaborative projects	The PI has previous experience in 3 EU collaborative projects and in a bilateral Germany-Spain collaborative project. The Co-PI has experience in 1 EU collaborative project, in a bilateral Germany-Spain collaborative project, and in 3 USA projects (NIH grants)
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
1) Krstić L, Vallejo R, Rodriguez-Rojo S, González-García MJ, Arias FJ, Girotti A, Diebold Y. Effective ocular delivery of antioxidant polyphenols using elastin-like polymer nanosystems developed by sustainable process. Int J Pharm. 2025 Jun 10;678:125691. doi: 10.1016/j.ijpharm.2025.125691. *Impact Factor = 5.2; Position Journal/Field: 43/352 Pharmacology & Pharmacy (Q1)* *2) Diebold Y, García-Posadas L. Ex vivo applications of porcine ocular surface tissues: Advancing eye research and alternatives to animal studies. Histol Histopathol. 2025 Aug;40(8):1139-1151. doi: 10.14670/HH-18-873.* *Impact Factor = 2; Position Journal/Field: 45/90 Pathology (Q2)* *3) Romero-Castillo I, López-García A, Diebold Y, García-Posadas L. Enrichment protocols for human conjunctival extracellular vesicles and their characterization. Sci Rep. 2024 Nov 16;14(1):28270. doi: 10.1038/s41598-024-79481-1.* *Impact Factor = 3.9; Position Journal/Field: 25/136 Multidisciplinary Science (Q1)*

4) García-Posadas L, Romero-Castillo I, Brennan K, Mc Gee MM, Blanco-Fernández A, Diebold Y. Isolation and Characterization of Human Conjunctival Mesenchymal Stromal Cells and Their Extracellular Vesicles. Invest Ophthalmol Vis Sci. 2023 Sep 1;64(12):38. doi: 10.1167/iovs.64.12.38.
Impact Factor = 4.4; Position Journal/Field: 7/62 Ophthalmology (Q1)
5) Diebold Y, García-Posadas L. Is the Conjunctiva a Potential Target for Advanced Therapy Medicinal Products? Pharmaceutics. 2021 Jul 26;13(8):1140. doi: 10.3390/pharmaceutics13081140.
Impact Factor = 6.525; Position Journal/Field: 39/279 Pharmacology & Pharmacy (Q1)

Up to 5 relevant previous projects, grants, networks or collaborations
Only projects in which I was local PI and/or coordinator: 1. *Tailoring nanoparticle-extracellular vesicle association as therapeutic platform for selective ocular surface tissue targeting (NANOSEVs). Ministry of Science and Innovation, PID2023-148252OB-C21 (Coordinated), National with international collaborators (225,000€), 2024/2028.* 2. *Human conjunctival 3D model for investigating the ocular effects of biological therapy (3D-Conj-OSD). Private international project (164,884€), 2024-2027.* 3. *Ocular topical administration of sorbitan ester nanoparticles for intraocular neuroprotection (RETINARTEX). Ministry of Science and Innovation, PDC2022-133239-C21 (Coordinated), National (115,000€), 2022/2024.* 4. *Delivery of artificial exosomes to treat blinding eye diseases (ARTEX4EYE). Ministry of Science, Innovation and University, RTI2018-094071-B-C21 (Coordinated), National with international collaborators (217,800€), 2019/2022.* 5. *Drug Discovery and Delivery NEtwork for ONcology and Eye Therapeutics (3D NEONET). H2020-MSCA-RISE-2016, Grant Agreement nº. 734907, International (945,000€ (UVA 37.600€)), 2017/2022.* 6. *Drug Discovery and Development of Novel Eye Therapeutics (3D-NET). FP7-PEOPLE-2013-IAPP, Marie Curie Industry-Academia Pathways and Partnerships (IAPP) action, Grant Agreement nº. 612218, Internacional (1,663,674€ (UVA 34,967.12€)), 2013/2017.*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers X Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
• *Experience in isolating, characterizing and functional studies of ocular surface cell-derived extracellular vesicles (only team in IOBA-UVa with such a strength).* • *Strong know-how in 3Rs compliant in vitro and ex vivo models to study ocular surface pathophysiology.* • *Experience with diverse nanotechnological platforms to develop ocular surface nanomedicines.*

Capabilities or expertise you would especially like to access through the consortium
• *Biomaterial experts and methods to improve our engineered 3D models of ocular surface structures.* • *Translational pathways to achieve extracellular vesicle-based therapies.* • *Expertise in different drug delivery-based approaches to develop regenerative and/or antiinflammatoy nanomedicines.*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X **Understood**

Name: Yolanda Diebold
Date completed: 22 May, 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	https://www.ibgm.uva.es/investigaci%C3%B3n/grupos-de-investigaci%C3%B3n/calcio-y-funci%C3%B3n-celular
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Sanz y Forés 3, 47003 Valladolid, Spain
Department(s), institute(s), research group(s) or unit(s) that would be involved	Calcium and Cellular Function Group, Institute for Biomedicine and Molecular Genetics (IBGM)
Additional institutional resources relevant to collaboration	Facilities and equipment's of the IBGM-UVa (https://www.ibgm.uva.es/*and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Maria Teresa Alonso, Full professor, Departemnt of Biochemistry, Molecular Biology and Physiology
Main contact email	talonso@uva.es

Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Advanced imaging technologies

Free Keywords for internal searchable database (please provide up to 10)	calcium imaging; live-cell imaging; Ca2+ biosensors; luminescence; organelle crosstalk; aequorin; transgenics; aging; neurodegeneration; Drosophila and mouse models.
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
Our research has focused on the role of intracellular calcium as a signaling molecule in various cellular paradigms and models, both in health and disease. Our group was a pioneer in Spain in microfluorescence and Ca2+ imaging in living cells, with single cell resolution, generating indicators, protocols, infrastructures and methodologies for monitoring Ca2+ (know-how) that has gained us an international reputation in the calcium signaling field. In the last two decades our group has focused on the role of the cytoplasmic organelles in Ca2+ signaling, especially mitochondria, endoplasmic reticulum and nucleus, and recently the endo-lysosome. To address this, we have developed a number of novel tools based on the bioluminescent Ca2+ binding protein aequorin, which have provided highly novel insights in the Ca2+ signaling field. For instance, we have demonstrated that chromaffin cells exhibit a very specialized calcium mechanism (CICR) typical of heart cells (JCB, 1999) and reported for the first time that mitochondria can take up unexpectedly high amounts of Ca2+ in the millimolar range under physiological conditions (Nat Cell Biol, 2000) or reported mitochondrial (JBC, 2001) and nuclear calcium oscillations (JBC, 2002). We have contributed to the role of organelles in neurosecretory cells (chromaffin and pituitary) as well as primary cerebellar granule cells. We have also generated a new family of fluorescent calcium genetic indicators optimized for organelles (PNAS, 2014; Cell Chem Biol, 2016). Our research extends from a molecular level in vitro, using the purified protein to explore mechanistic aspects of the structure-function relationship, up to level of the complete organism, with the generation of transgenic animals and their application in in vivo studies in the whole animal in physio-pathological aspects such as aging (JCS, 2020) or neurodegeneration in models such as Huntington´s disease.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Because Intracellular calcium acts as a ubiquitous second messenger across almost all living cells, measuring intracellular Ca2+ dynamics provides a direct readout for several distinct processes. Ca2+ signalling is a primary proxy for cellular activity, metabolic demand and cellular excitability. It can be widely applied in very different systems or models such as drug screening, live-cell functional analysis and neurodegenerative disease modelling. Therefore, the specific methodologies provided by the group could support translational eye-health projects focused on retinal neurodegeneration, cellular stress signalling and biomarker discovery.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.*
- live-cell calcium imaging
- Intraorganellar calcium imaging
- biosensor development
- image analysis
- microfluorescence
- intracellular signaling
- luminal endo-lysosomal calcium dynamics
- transgenic Drosophila and mouse disease models
- neurodegeneration research

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

Advanced microscopy and microfluorescence platforms; luminescent equipments; genetically encoded calcium sensors; live-cell imaging systems; experimental transgenic Drosophila and mouse models;

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Training in calcium imaging methodologies, generation of Drosophila lines, in vivo imaging of intraorganellar calcium, biosensor use, image analysis, functional live-cell assays and intracellular signalling studies

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Access to advanced retinal disease models, collaborations in translational neuroscience and ophthalmology, integration of calcium signalling approaches into retinal and neurodegenerative disease research, application of the calcium technologies to the ophthalmology field.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
María Teresa Alonso	Full Professor	Biochemistry	Researcher	Experienced	talonso@uva.es
Thomas Schimmang	National Research Council (CSIC) Investigator	Developmental Biology	Researcher	Experienced	thomas.schimmang@uva.es
Jonathan Rojo	Senior Researcher	Nursing	Researcher	Experienced	jonathan.rojo@uva.es
Leonardo Gallegos	Postdoctoral Researcher	Biochemistry	Researcher	Experienced	leonardo.gallegos@uva.es
Beatriz Callejo	PhD student	Biochemistry	Researcher	Early stage	beatriz.callejo@uva.es
Cynthia Sánchez	PhD student	Biochemistry	Researcher	Early stage	cynthia.sanchez@uva.es

Main disciplines represented in your team	physiology, molecular biology, cell biology, genetics,
Experience in EU / international / multi-partner collaborative projects	M.T.A. participated as a researcher in the project "Neuron -Disturbance of calcium processing by intracellular organelles in the ER-mitochondria calcium cycle in a triple mutant model of Alzheimer disease." Ref.: ERA-net NEURON - SAF2008-03175-E. Funding Agency: European Commission/MICINN. Position: P.I. Spanish partner, University of Valladolid (J. García-Sancho). Start: 2009 – End: 2012. Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note: Experience with stem cell models, viral gene transfer, CRISPR/Cas9-mediated gene modification

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

1. Calvo, B., Torres-Vidal, P., Delrio-Lorenzo, A., Rodriguez, C., Aulestia, F.J., Rojo-Ruiz, J., Callejo, B., McVeigh, B.M., Keller, M., Grimm, C., Oorschot, V., Moiseenkova-Bell, V., Yule, D.I., Garcia-Sancho, J., Patel, S. and Alonso, MT*. (2025). Direct measurements of luminal Ca2+ with endo-lysosomal GFP-aequorin (ELGA) reveal functional IP3 receptors. J.Cell Biol. 224. We provide a novel tool to measure luminal endo-lysosomal Ca2+ and novel insights into the mechanisms governing Ca2+ handling in the endo-lysosomal system. This work has been selected for the Spotlight section in the same issue (Morgan, AJ (2025) J Cell Biol 224). Also, this work is the main body of P. Torres´s PhD thesis.

2. Delrio-Lorenzo, A., Rojo-Ruiz, J., Torres-Vidal, P., Alonso, M.T*., and García-Sancho, J. (2024). In vitro and in vivo calibration of low affinity genetic Ca2+ indicators. Cell Calcium 117, 102819. We described here a novel calibration procedure to accurately convert the signal of the fluorescent indicator in high Ca2+, like the endoplasmic reticulum (ER) into [Ca2+]ER, both in vitro and in vivo (in erGAP3 transgenic flies). This work is part of the PhD thesis of A. Del Río.

3. Gómez-Valadés AG, Pozo M, Varela L, … Alonso MT (18/26), …, Claret M. (2021) Mitochondrial cristae-remodeling protein OPA1 in POMC neurons couples Ca2+ homeostasis with adipose tissue lipolysis. Cell Metab. 33: 1820-1835. Dr. Claret contacted us as experts in mitocondrial Ca2+ to image mitocondrial Ca2+ specifically in POMC neurons in acute hypothalamus slices of KO-OPA1 mice.

4. Rojo-Ruiz, J., Navas-Navarro, P., Nuñez, L., García-Sancho, J., and Alonso, MT*. (2020) Imaging of Endoplasmic Reticulum Ca2+ in the Intact Pituitary Gland of Transgenic Mice Expressing a Low Affinity Ca2+ Indicator. Frontiers in Endocrinology 11: 615777. doi: 10.3389/fendo. 615777. In this article we showed that the erGAP3 transgenic mice show robust functional expression in pituitary gland where we have simultaneously studied the Ca2+ signals evoked by various releasing hormones in the intact gland. We demonstrated unambigously the existance of polireceptorial cells that respond to several hormones. This work was part of the P. Navas's PhD thesis.

5. Delrio-Lorenzo A, Rojo-Ruiz J, Alonso MT, García-Sancho J. (2020) Sarcoplasmic reticulum Ca2+ decreases with age and correlates with the decline in muscle function in Drosophila. J Cell Sci. 133: jcs240879. In this work we tested the hypothesis that the age-associated loss of mass and muscle force (sarcopenia) has its origin in the loss of Ca2+ homeostasis in the sarcoplasmic reticulum (SR). We indeed could demonstrate that aged flies stored 10-fold less Ca2+ in the SR of thoracic muscles than young flies and this correlated well with the process of sarcopenia. This work was the main body of A. del Río's PhD thesis.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). *Indicate if international.*

1. PID2023-146434NB-I00. Physiological and molecular basis of endo-lysosomal Ca2+ dynamics in health and disease. DGI MICINN. P.I. MTA. Start: 01/01/2025 – End: 31/8/2027. 325.000€.

2. PID2020-116086RB-I00. Calcium and Cellular Function. DGI MICINN. P.I. MTA. (Co-PI: J. García-Sancho). Start: 01/01/2022 – End: 31/8/2024. (extended until 31/12/2024) 340.000€.

3. BFU 2017-83066-P. Calcium and Cellular Function. DGI MICINN. P.I. MTA (Co-PI: J. García-Sancho). Start: 01/01/2018 – 31/12/2020 (extended until 31/12/2021). 363.000€.

4. BFU2014-53469-P. Calcio y Función Celular. DGI MICINN. Position: Co-P.I. (P.I. - J. García-Sancho). 01/01/2015 – 31/12/2017. 423.500€.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

1. P201230475. M.T. Alonso and J. García-Sancho Martín. Calcium sensors and methods for the detection of intracellular free calcium. Country of priority: Spain. 29.03.2012. University of Valladolid

2. P201231104. M.T. Alonso and J. García-Sancho Martín. Apoaequorin mutants and methods for their use. Country of priority: Spain. 13.07.2013. University of Valladolid.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers X Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	X Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. Calcium imaging, intraorganellar calcium measurements, live-cell biosensors, in vivo functional assays and neurodegenerative disease models

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Advanced retinal disease models, translational ophthalmology collaborations, AI-based image analysis and multimodal neurodegeneration approaches

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Maria Teresa Alonso
Date completed: 22/9/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	UNIVERSIDAD DE VALLADOLID
Short name / acronym used in projects	UNIVERSIDAD DE VALLADOLID
Participant Identification Code (PIC), if available	999862809
Website(s)	https://innograinlab.uva.es/ https://isp.uva.es/en/
Country	Spain
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Edificio I+D, Campus Miguel Delibes, Paseo de Belén 11, 47011, Valladolid
Department(s), institute(s), research group(s) or unit(s) that would be involved	InnograinLab Research Group (GIR 464, Bioquímica y Estructura de Moléculas y Matrices Alimentarias), Institute of Sustainable Processes
Additional institutional resources relevant to collaboration	e.g. core facilities, clinical units, technology platforms, animal facilities, data infrastructures

Main contact person for this profile	Mario Martinez Martinez, Department of Agricultural Engineering
Main contact email	mmartinez@uva.es
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): Analysis of structural and colloidal properties of bio-based polymers, metabolimics (profiling of bioactive compounds), advanced in vitro digestion and colonic fermentation models, extraction and processing of plant-based biopolymers.

Free Keywords for internal searchable database (please provide up to 10)	Precision nutrition, In vitro digestion, Microbiota, Metabolomics, Polymer structure, Polysacharides, Polyphenols, Glycomics, Compounds extraction, Colloidal properties
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
We are a leading group focused on designing and utilizing plant macromolecule assemblies that precisely target metabolic responses, bioactivity, and technological performance for applications in food, agriculture, pharmaceuticals, and packaging.

Why your team could be relevant to a translational eye-health and vision consortium
Our group could contribute through our expertise in designing plant-based macromolecular assemblies that modulate metabolic responses, improve bioactive delivery, and enhance material functionality. This knowledge could support the development of sustainable biomaterials and delivery systems for ocular health applications, including antioxidants and anti-inflammatory compounds relevant to retinal and eye diseases. In addition, our experience in digestion, bioaccessibility, and exposomic interactions could help investigate how nutrition, metabolism, and bioactive compounds influence eye health and disease progression. We are also experts in extraction and processing techniques, which could be harnessed to obtain and design delivery systems or bio-based molecules.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
- Increase the precision in understanding the relationships between food structure, metabolic response, and the gut microbiome. *- Structure-function relationships in relation to building novel foods and packaging materials.* *- Analytical methods to elucidate structural features of biopolymers, including monomer composition, linkage patterns, sequencing, hydrodynamic and molecular weight, anomeric configuration and chain length distributions.* *- Chemical, enzymatic and physical modifications of plant polysaccharides and proteins.* *- Interactions between food macromolecules and small metabolites (e.g., polyphenols) in complex systems.*

- Health aspects, particularly the digestion and bioaccesibility of plant biopolymers and phenolic compounds and the role of polysaccharides as dietary fiber.
- Physical properties, including the rheological and colloidal properties of edible systems.
- The encapsulation and controlled release of active compounds for inclusion in food formulations.
- The film forming properties of plant polysaccharides and proteins with application in edible films, coatings and active packaging.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Our research infrastructure can be found in our website - https://innograinlab.uva.es/research-infrastructure/.
The group holds specialized expertise and facilities including advanced analytical equipment; molecular biology and omics resources; BSL-2 microbiology lab; static and semi-dynamic INFOGEST in vitro digestion setups; and anaerobic digesters/fermenters (2–200 L). Additionally, we have access to established agreements with specialized partners for high-throughput microbiome sequencing.
In particular, the research team laboratory locate offers two IC-PAD (850 Professional IC, Metrohm and Dionex ICS-6000, ThermoFisher Scientific), a SEC-FFF-UV-MALS-RI-IV (Agilent 1260, Wyatt Eclipse, DAWN, ViscoStar and Optilab, Agilent-Wyatt), a Fourier transformed infrared Spectrometer (ATR-FTIR Nicolet iS50, ThermoFisher), two 500 MHz solid/liquid state NMR (Agilent and Bruker), a miniSpec low field NMR (Bruker), a 6 Flex Real-Time PCR System (QuantStudio), Powder X-ray diffractometer (XRD, Bruker D8 Discover A25), an Environmental scanning electron microscope (ESEM, Quanta 200FEI), a confocal laser scanning microscope (CLSM, Leica TCS SP8X), a DLS Zetasizer Advance Pro Red Label (Malvern) and a Mastersizer (Malvern). In addition, the research team laboratory is also equipped with a UPLC-ESI-QTOF-MS (Maxis Impact, Bruker), a GC-MS (Agilent) and a MALDI-TOF-MS (Autoflex, Bruker). The research team counts with regular sample preparation and conventional microbiology labs equipped with ThermoMixer with thermoblock reaction vessels (Eppendorf F2.0); Ultraturrax homogenizer (IKA-T18), moisture analyzer (MA 150 Sartorius), Muffle (Buhler), Nitrogen-evaporator (N-EVAP, Organomation), ultrasonic water bath, incubators, laminar flow cabinets, freeze-drier, stirring plates, shaking and thermostatic water baths, centrifuges, pH meter, and several plate spectrophotometric readers. InnograinLab also count with a controlled-stress oscillatory rheometer (DHR20, TA Instruments), Differential scanning calorimeter, (DSC Q-20, TA instrument), spectrocolorimeter (Minolta CN-508i) and Texture analyzer (TA-XT2, Stable Microsystems).

Training capacity and knowledge-transfer potential

Upcoming visiting researchers would benefit from interdisciplinary training combining advanced food, biomaterial, microbiology, and analytical sciences with translational applications in health and sustainability. Training opportunities could include hands-on experience in advanced characterization techniques such as SEC-FFF-MALS, NMR, FTIR, XRD, CLSM, DLS, rheology, thermal analysis, and mass spectrometry platforms (UPLC-QTOF-MS, GC-MS, MALDI-TOF-MS), as well as molecular biology and omics-related workflows. Researchers could also receive practical training in static and semi-dynamic INFOGEST in vitro digestion models, fermentation systems and anaerobic bioreactors, microbiology under BSL-2 conditions, and microbiome-oriented studies. In addition, the group can provide expertise in plant macromolecule assemblies, food structure–function relationships, bioaccessibility and bioactivity assessment, colloidal systems, sustainable biomaterials, and data integration across physicochemical, biological, and metabolic responses. The infrastructure and multidisciplinary environment further support training in experimental design, method development, advanced analytical interpretation, and collaborative translational research.

What your team would hope to gain from participating

Access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, joint publications and longer-term strategic collaboration.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Mario Martinez Martinez	Profesor Titular	Department of Agricultural Engineering / Institute of Sustainable Processes	PTUN	A	mmartinez@uva.es
Laura Roman Rivas	Ramon & Cajal Researcher	Department of Agricultural Engineering /	Ramon & Cajal Researcher	B	lroman@uva.es

		Institute of Sustainable Processes			
Cristina Andrés Iglesias	Profesora Ayudante Doctor	Department of Agricultural Engineering / Institute of Sustainable Processes	Profesora Ayudante Doctor	C	cristina.andres.iglesias@uva.es

Main disciplines represented in your team	Biopolymer chemistry, Food chemistry, Applied biotechnology, Metabolomics, Material properties
Experience in EU / international / multi-partner collaborative projects	Mario Martínez is the Coordinator of the FutureFoods partnership co-funded by the European Union's Horizon Europe Programme under Grant Agreement 101136361 with a total funding of 1.2 M euro. Mario Martínez is the Host Principal Investigator of two MSCA-PF. Mario Martínez is the Principal Investigator of one MSCA-SE led by the University of Aveiro (Portugal). Mario Martínez coordinated the submission of two EIC Pathfinder proposals requesting 7 M euro in total that received good grades but were not funded. Laura Roman is the principal investigator of a Interreg project led by the Polytech Instituto of Braganca (Portugal).
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. *Bechtner, J. D., Lopez-Rodulfo, I. M., Gallego-Lobillo, P., Hosek, J., Schwab, C., & Martinez, M. M. (2025). Polyphenols non-covalently bound to artificial plant cell walls are readily accessible to fecal microbiota and transiently alter fermentation activity. Food Chemistry, 145817.*
2. *Lopez-Rodulfo, I. M., Rasmussen, M. K., Hamaker, B., & Martinez, M. M. (2026). Selective detoxification of digesta revealed how cold-pressed apple fractions modulate transepithelial glucose transport and stimulate GLP-1 secretion. Food Chemistry, 148588.*
3. *Vidal, N. P., Charlampita, M. C., Spotti, M. J., & Martinez, M. M. (2024). Multifunctional phloroglucinol-loaded pea starch coating for refrigerated salmon. Food Packaging and Shelf Life, 43, 101277.*
4. *Roman, L., Baylis, B., Klinger, K., de Jong, J., Dutcher, J. R., & Martinez, M. M. (2022). Changes to fine structure, size and mechanical modulus of phytoglycogen nanoparticles subjected to high-shear extrusion. Carbohydrate Polymers, 298, 120080.*
5. *Roman, L., Yee, J., Hayes, A. M., Hamaker, B. R., Bertoft, E., & Martinez, M. M. (2020). On the role of the internal chain length distribution of amylopectins during retrogradation: Double helix lateral aggregation and slow digestibility. Carbohydrate Polymers, 246, 116633.*

Up to 5 relevant previous projects, grants, networks or collaborations

1. *1st FutureFoodS co-funded call. Biotechnological upcycling of fruit pomace into prebiotic food ingredients (BIOPOM). Horizon Europe Framework Program and Spanish Ministry of Science and Innovation & European Union. 01/04/2026 - 01/04/2029. Amount awarded: 1,291,990.00 € (gestionado como Project Coordinator, Mario Martinez) PI: Mario Martinez (University of Valladolid). Role: Project coordinator and Principal investigator. Other PIs, Clarissa Schwab (Aarhus University), Lillian Barros (Polytechnique Institute of Braganca) and Magdalena Svanström (Chalmers University of Technology).*
2. *Marie Sklodowska Curie Actions-Postdoctoral Fellowships 2025. Decoding Ileal Metabolic Responses to Complex Dietary Structures for Precision Nutrition (PREILEAL). Horizon Europe Framework Programme (HORIZON). PIs: Ivan Misael Lopez Rodulfo & Mario Martinez. Role: co-Principal Investigator/Coordinator at host institution. Importe: 209.914 €. Ámbito europeo.*
3. *Danmarks Frie Forskningsfond - Sapere Aude 2021. 1051-00046B Non-covalent Binding to Target Prime Sites for Cardiometabolic Protection. Independent Research Fund Denmark (DFF). 01/01/2022-01/01/2026. PI: Mario Martinez. Importe: 833.511 €. Ámbito nacional.*

4. *Proyectos de Generación de Conocimiento 2024. Green crosslinking of bread waste particles for food integration and precision gut microbiome modulation (BREBIOTICS). Agencia Estatal de Investigación (AEI). 01/01/2024-31/12/2027. Amount awarded: 351.700€, PI: Mario Martinez (University of Valladolid). Role: Principal investigator.*
5. *Consolidación Investigadora 2023. Bio-inspired food packaging: harnessing plant cell wall waste infused with vacuolar polyenols (WASTEtoPACK). Spanish Ministry of Science and Innovation & European Union. 01/04/2024-01/04/2026. Amount awarded: 200.000 €. PI: Mario Martinez (University of Valladolid). Role: Principal investigator.*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers x Technical Staff/Innovation x Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
We are a leading group focused on designing and utilizing plant macromolecule assemblies that precisely target metabolic responses, bioactivity, and technological performance for applications in food, agriculture, pharmaceuticals, and packaging.

Capabilities or expertise you would especially like to access through the consortium
Upcoming visiting researchers would benefit from interdisciplinary training combining advanced food, biomaterial, microbiology, and analytical sciences with translational applications in health and sustainability. Training opportunities could include hands-on experience in advanced characterization techniques such as SEC-FFF-MALS, NMR, FTIR, XRD, CLSM, DLS, rheology, thermal analysis, and mass spectrometry platforms (UPLC-QTOF-MS, GC-MS, MALDI-TOF-MS), as well as molecular biology and omics-related workflows. Researchers could also receive practical training in static and semi-dynamic INFOGEST in vitro digestion models, fermentation systems and anaerobic bioreactors, microbiology under BSL-2 conditions, and microbiome-oriented studies. In addition, the group can provide expertise in plant macromolecule assemblies, food structure–function relationships, bioaccessibility and bioactivity assessment, colloidal systems, sustainable biomaterials, and data integration across physicochemical, biological, and metabolic responses. The infrastructure and multidisciplinary environment further support training in experimental design, method development, advanced analytical interpretation, and collaborative translational research.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

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☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	UNIVERSIDAD DE VALLADOLID
Short name / acronym used in projects	UVA
Participant Identification Code (PIC), if available	999862809
Website(s)	https://presstech.uva.es/en/ Pressure Technologies Group (PressTech) https://bioeconomia.uva.es/en/ Institute of Bioeconomy (BioEcoUVa)
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public bodv ☐ Other
Main address	Prado de la Magdalena 3-5, Escuela de Ingenierías Industriales, 47011, Valladolid, España
Department(s), institute(s), research group(s) or unit(s) that would be involved	Pressure Technologies Group (PressTech) of the Institute of Bioeconomy (BioEcoUVa) at the University of Valladolid
Additional institutional resources relevant to collaboration	Lab PressTech of the Pressure Technologies Group https://presstech.uva.es/en/labs-and-equipment/

Main contact person for this profile	Juan García Serna, Full Professor at the Department of Chemical Engineering and Environmental Technology of the University of Valladolid

Main contact email	juan.garcia.serna@uva.es
Other contact person(s), if relevant	María José Cocero Alonso, Full Professor at the Department of Chemical Engineering and Environmental Technology of the University of Valladolid, Director of PressTech Research Group and Director of Institute of Bioeconomy.

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Analysis of the components in the lab, extraction or reaction. We can also create digital tools using AI.

Free Keywords for internal searchable database (please provide up to 10)	perfluorooctane (PFO), impurity identification (GC-MS), chemical purification, process scale-up, chemical process engineering, supercritical fluid technology, hydrothermal processing, green chemistry and sustainable processes, bioactive compound extraction from biomass, catalysis and reaction engineering
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
Our team belongs to the Pressure Technologies Group (PressTech) at the Institute of Bioeconomy (BioEcoUVa) formerly High Pressure Processes Group, University of Valladolid. We work in chemical engineering and sustainable chemistry, designing, modelling and scaling up sustainable chemical processes, from laboratory reactors to pilot plant. *Our most direct link to eye health is our long-standing collaboration with the University of Valladolid Retina Group (IOBA) on the safety of perfluorocarbon liquids (PFCLs) used in vitreoretinal surgery. Following cases of severe acute retinal toxicity linked to contaminated batches of perfluorooctane (PFO), we identified and characterized the toxic impurities in commercial PFO using GC-MS and complementary analytical techniques. This work contributed to peer-reviewed studies on the chemical causes of intraocular toxicity (Regulatory Toxicology and Pharmacology, 2020; Scientific Reports, 2021). We then developed a chemical reaction and purification process to remove these impurities and scaled it up to pilot plant, showing that ophthalmic-grade liquids can be produced reproducibly and safely.* *Our broader expertise covers supercritical fluid technology, hydrothermal processing, catalysis and reaction engineering, and the extraction of bioactive compounds from biomass (e.g. ferulic acid, carotenoids) and biopolymers (pectins, hemicellulose and cellulose), including their formulation in liposomes. We also have strong expertise in process simulation with specialized process simulation software (Aspen Plus and Aspen HYSYS), sustainability assessment, life cycle assessment (LCA) and the development of artificial intelligence tools for process modelling and decision-making. Together, these capabilities allow us to contribute high-purity materials,*

analytical quality control, process scale-up and sustainability evaluation to translational ophthalmic research.

Why your team could be relevant to a translational eye-health and vision consortium

PressTech team offers a process engineering and green chemistry perspective that is rarely found in ophthalmology consortia but is often decisive for moving an innovation from the laboratory to the patient: ensuring that the materials, formulations and products used in the eye can be manufactured with the required purity, safety, reproducibility and sustainability.

Our work with the University of Valladolid retina group (IOBA) on perfluorocarbon liquids shows this added value in practice. After cases of severe acute retinal toxicity caused by contaminated perfluorooctane (PFO), we identified the toxic impurities in commercial PFO and silicone oils by GC-MS, developed a chemical reaction and purification route to remove them and scaled the process up to pilot plant. This scale-up and process expertise, can be applied to other materials complementing the consortium's.

Beyond this direct experience, our expertise in supercritical fluids and hydrothermal processing opens less conventional routes to innovation. These include solvent-free purification and extraction, supercritical CO_2-assisted processing of materials and devices, and the recovery of natural bioactive compounds with antioxidant properties (e.g. ferulic acid, carotenoids), which can be formulated in lipid-based carriers such as liposomes for potential delivery.

We also bring strong computational and sustainability capabilities: process simulation, techno-economic analysis, life cycle assessment (LCA) and artificial intelligence tools for process modelling and decision-making. These can help partners anticipate the cost, scalability and environmental footprint of new therapies and products from an early stage.

Our work is supported by PressTech lab, a 500 m² laboratory with high-pressure and pilot-scale equipment, and by the wider Institute of Bioeconomy (BioEcoUVa), which provides additional scientific and technical support. Altogether, our team strengthens the consortium's ability to turn promising ophthalmic research into safe, high-quality and sustainably manufactured solutions for patients.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

End-to-end chemical process development, from reaction and separation design at laboratory scale to pilot-plant validation. Our strong orientation towards industrial application is supported by extensive experience in competitive research projects, collaboration with industry and technology transfer.

- ***Chemical safety of ophthalmic materials:*** *identification and quantification of toxic impurities in perfluorocarbon liquids (e.g. perfluorooctane) and other components used in vitreoretinal surgery, by GC-MS and complementary analytical techniques, in collaboration with the IOBA retina group.*
- ***Purification and reaction engineering:*** *design of chemical reactions and purification routes to remove harmful impurities and obtain high-purity, ophthalmic-grade liquids and materials.*
- ***Process scale-up:*** *transfer of purification and production processes from laboratory to pilot plant, ensuring reproducibility, quality and readiness for industrial manufacturing.*
- ***Supercritical fluid technology:*** *supercritical CO_2 and water processes for solvent-free extraction, purification, material processing and particle or material engineering.*
- ***Hydrothermal processing and biorefinery:*** *recovery of high-value bioactive compounds with antioxidant properties from biomass (e.g. ferulic acid, carotenoids, polysaccharides such as pectins, hemicelluloses and celluloses). These compounds have been incorporated into lipid-based carriers such as liposomes in collaboration with formulation partners.*
- ***Catalysis and reaction engineering:*** *development and kinetic study of heterogeneous catalysts and continuous reactors for selective chemical transformations.*
- ***Process simulation and techno-economic analysis:*** *modelling of complete processes with specialized simulation software to estimate cost, scalability and manufacturing feasibility at an early stage.*
- ***Sustainability and life cycle assessment (LCA):*** *environmental footprint evaluation of processes and products using tailor-made sustainability indexes, supporting greener manufacturing of chemical products.*
- ***Artificial intelligence tools:*** *data-driven models and AI-based tools for process modelling, optimization and decision-making.*
- ***Integrated research ecosystem:*** *PressTech lab, a 500 m² facility with high-pressure and pilot-scale equipment, supported by the Institute of Bioeconomy (BioEcoUVa).*
- ***Collaborative flexibility:*** *ability to adapt our analytical, process and computational capabilities to consortium needs, from troubleshooting the chemical safety of specific products to designing scalable and sustainable manufacturing routes.*
- ***Soft skills:*** *a creative, problem-solving mindset with sustainability at the core of our work. We have a strong commitment to science communication and outreach to society, and experience in interdisciplinary teamwork with industry.*

Infrastructure, equipment, facilities or datasets relevant to consortium work

Facilities and equipment of 500 m² facility PressTech lab at the Chemical Engineering and Environmental Technology Department, School of Engineering, Institute of Bioeconomy (BioEcoUVa), University of Valladolid (https://presstech.uva.es/en/labs-and-equipment/).

*• **Analysis and characterisation lab:** GC-MS systems (Agilent), HPLC with PDA, IR and UV detectors (Waters), portable micro-GC, FT-IR (Bruker ALPHA), portable Raman spectrometer, TOC analyser, particle size analyser (Malvern Mastersizer 2000), surface tension meter (Krüss), moisture analyzer, among others. This lab is the basis for impurity identification and quality control of ophthalmic liquids and materials.*

*• **High-pressure view cells:** four sapphire-window cells (Eurotechnica, SITEC; up to 500 °C and 500 bar) with video recording, for studying phase behaviour, solubility and reactions in supercritical fluids.*

*• **Formulation and particle engineering:** supercritical antisolvent (SAS) pilot plant for microparticles and microencapsulation, PGSS particle generation and drying pilot plant, supercritical CO_2 aerogel drying unit, emulsion formulation plant, spray dryer (Niro Mobile Minor), freeze-dryer, high-shear homogeniser (IKA) and dissolution testing apparatus (Copley, 6 positions).*

*• **Reaction and extraction units:** continuous hydrothermal reactor (up to 454 °C and 225 bar), continuous supercritical CO_2 reaction plant, batch hydrogenation reactor (Berghof), supercritical fluid extraction unit (5 L, up to 400 bar), microwave reactors and synthesisers (Anton Paar, CEM), tubular furnace and inert-atmosphere glovebox.*

*• **Pilot plants (TRL 3–5):** pressurised hydrolysis units PHUn 1 and PHUn 2 (0.5–5 kg/h biomass), patented multi-bed pilot plant PHUn 0, supercritical CO_2 extraction pilot plant and continuous microwave-assisted extraction pilot plant. These plants allow processes to be scaled from laboratory to pilot scale.*

*• **Sample preparation:** planetary ball mill, blade mill, wet mill, filter press and rotary evaporators.*

*• **Computational tools:** Aspen Plus and Aspen HYSYS process simulation software, SimaPro life cycle assessment (LCA) tools and in-house AI and data-driven modelling tools for process design, optimization and sustainability assessment. AI coding tools.*

Training capacity and knowledge-transfer potential

Inbound staff would join a highly interdisciplinary training environment in chemical engineering, green chemistry and process technology within PressTech lab, one of Europe's leading groups in supercritical fluids and high-pressure technologies, supported by the Institute of Bioeconomy (BioEcoUVa) of the University of Valladolid.

From a scientific and technical perspective, visitors would gain hands-on experience in the chemical analysis and quality control of materials and liquids, including impurity identification by GC-MS, HPLC, FT-IR and Raman spectroscopy. This expertise has been applied to perfluorocarbon liquids and other products used in vitreoretinal surgery. They would also be trained in reaction engineering, supercritical fluid and hydrothermal processing, particle engineering and microencapsulation, and the extraction of bioactive compounds from natural sources. A key added value is the opportunity to take a material, formulation or process developed at their home institution and test its purification, formulation or scale-up in our facilities, from bench scale to pilot plants at TRL 3–5.

Inbound researchers could also be trained in process simulation, techno-economic analysis, life cycle assessment (LCA) and artificial intelligence tools for process modelling and optimization. These give clinical and biomedical researchers a practical view of how new ophthalmic products can be manufactured safely, reproducibly and sustainably.

On the translational and entrepreneurial side, staff would benefit from our long-standing collaboration with industry, patent co-invention and industrial R&D experience (which includes six years at Renmatix, Inc. (USA), or two years in Tecnicas Reunidas Madrid). Training would combine lab work with group seminars, doctoral school activities of ESDUVa (soft and transversal skills), international courses of the EFCE High Pressure Chemical Engineering working group and conferences.

Our track record supports this capacity. Over the last decade our team has supervised numerous doctoral theses, many of them international or in cotutelle with Åbo Akademi University (Finland) and Hamburg University of Technology (Germany). Our doctoral graduates have over 95% employability in companies, technology centers and universities in Spain and abroad.

What your team would hope to gain from participating

As chemical engineers, we see participation in this consortium as an exceptional opportunity to learn. Our team is curious by nature and strongly motivated to apply our knowledge to challenges with a direct impact on society and patient health. We are especially interested in gaining insight into ophthalmology and vision science from the clinical and biological side: how ocular diseases are studied, how

therapies and medical devices are validated, and which unmet needs in materials, formulations and manufacturing currently limit their translation to patients.

In particular, we would like to gain access to preclinical models, biocompatibility and toxicity testing platforms, clinical perspectives and regulatory know-how. These would allow us to validate the chemical, formulation and process solutions we develop in a biologically and clinically meaningful context. We are also keen to learn from partners working in drug delivery, advanced therapies, data science and AI, and patient-oriented research. We believe interdisciplinary cross-fertilization between these fields and chemical engineering can open unexpected routes to innovation.

A central objective is to build long-term international collaborations that go beyond short-term exchanges and lead to joint projects, coordinated funding applications, patents and high-impact joint publications. We strongly support the mobility of our PhD students and early-career researchers. Exposure to biomedical and clinical environments would broaden their perspectives, strengthen their adaptability and international networks, and open new career paths at the interface between chemistry, engineering and health.

Finally, we are convinced of the key role of the chemical industry in health and well-being. According to the Spanish Chemical Industry Federation (FEIQUE), the chemical sector is the leading exporting sector of the Spanish economy, supports over one million direct, indirect and induced jobs and accounts for almost a quarter of all industrial R&D investment in Spain. Through this consortium, we aim to strengthen the connection between chemical engineering and eye health, contributing to safer, more sustainable and more accessible solutions for patients transferring knowledge and creating innovative ideas.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
JUAN GARCÍA SERNA	Full Professor	Chemical Engineering and Environmental Tech.	Researcher	R4 – Leading Researcher	juan.garcia.serna@uva.es
DANILO ALBERTO CANTERO SPOSETTI	Associate Professor (permanent position)	Chemical Engineering and Environmental Tech.	Researcher	R3 – Established Researcher	danilo.cantero@uva.es
FACUNDO MATTEA	Distinguished Senior Researcher Beatriz Galindo	Chemical Engineering and Environmental Tech.	Researcher	R3 – Established Researcher	facundo.mattea@uva.es
MARTA RAMOS ANDRÉS	Postdoctoral Fellowship UVa	Chemical Engineering and Environmental Tech.	Researcher	R2 - Recognized Researcher	marta.ramos@uva.es
VESNA LEONTIJEVIC	Postdoctoral Fellowship UVa	Chemical Engineering and Environmental Tech.	Researcher	R2 - Recognized Researcher	vesna.leontijevic@uva.es
ANDREA CASAS GONZÁLEZ	PhD student UVa and researcher hire by a project	Chemical Engineering and Environmental Tech.	Researcher	R1 – First Stage Researcher	andreapatricia.casas@uva.es
MIGUEL ÁNGEL PEÑALOZA RODRIGUEZ	PhD student UVa and researcher hire by a project	Chemical Engineering and Environmental Tech.	Researcher	R1 – First Stage Researcher	miguelangelpr@uva.es
To be defined (TBD) x1	PhD student UVa	Chemical Engineering and Environmental Tech.	Researcher	R1 – First Stage Researcher	N/D

Main disciplines represented in your team	Chemical engineering, process engineering and scale-up, green and sustainable chemistry, analytical chemistry, catalysis and reaction engineering, supercritical

	fluid technology, particle engineering and formulation, process simulation, sustainability and life cycle assessment (LCA), AI and data-driven modeling, and technology transfer.
Experience in EU / international / multi-partner collaborative projects	Extensive experience in EU-funded, national and industry-funded multi-partner projects on biorefineries, supercritical and hydrothermal processing and sustainable chemical processes. We also have long-standing international academic collaborations (e.g. joint PhDs with Åbo Akademi University, Finland) and industrial partnerships.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Coco-Martin R.M., Andrés-Iglesias C., Srivastava G.K., Lagos-Rodriguez J., Ruiz-Tevah M., Díaz-Cárdenas M.R., Fernandez-Bueno I., García-Serna J., García-Gutierrez M.T., García-Layana A., Pastor J.C. Intraocular toxicity caused by MEROCTANE perfluorocarbon liquid (2021) Scientific Reports, 11 (1), art. no. 599, Cited 11 times. DOI: 10.1038/s41598-020-79561-y
2. Srivastava G.K., Andrés-Iglesias C., Coco R.M., Fernandez-Bueno I., Medina J., García-Serna J., Dueñas A., Rull F., Pastor J.C. Chemical compounds causing severe acute toxicity in heavy liquids used for intraocular surgery (2020) Regulatory Toxicology and Pharmacology, 110, art. no. 104527, Cited 18 times. DOI: 10.1016/j.yrtph.2019.104527
3. Menalla E., Serna J.G., Cantero D., Cocero M.J. Hydrothermal hydrolysis of triglycerides: Tunable and intensified production of diglycerides, monoglycerides, and fatty acids (2024) Chemical Engineering Journal, 493, art. no. 152391, Cited 35 times. DOI: 10.1016/j.cej.2024.152391
4. Araujo-Barahona G., De Simone M., Brunberg C., Eränen K., Reinsdorf A., Roos M., García-Serna J., Russo V., Murzin D.Y., Salmi T. Solid raney-type Ni foam catalysts for xylitol production: Continuous and batch operation (2025) Applied Catalysis A: General, 701, art. no. 120324, Cited 4 times. DOI: 10.1016/j.apcata.2025.120324
5. García-Serna J., Piñero-Hernanz R., Durán-Martín D. Inspirational perspectives and principles on the use of catalysts to create sustainability (2022) Catalysis Today, 387, pp. 237 - 243, Cited 64 times. DOI: 10.1016/j.cattod.2021.11.021

Up to 5 relevant previous projects, grants, networks or collaborations

1. CPP2021-008695, JUSTPERFLUORO - DESIGN, DEVELOPMENT AND VALIDATION OF A HIGHLY PURIFIED PERFLUROOOCTANE FOR USE IN OPHTHALMIC SURGERY. State plan for scientific, technical and innovation research 2021-2023. Iván Fernández. (Agencia Estatal de Investigación; European Union -Next Generation EU; Recovery, Transformation and Resilience Plan). 01/02/2022-28/02/2025. 110.784 €.
2. Sustainable pulping innovation achieving zero pollution through efficient wood residue feedstock utilisation. 1/5/2025-30/4/2028, ref. 101180671, UVa=219750€, Horizon Europe, **Garcia Serna, Juan (IP UVa**). Total = 3,998,486.25€ coordinated by VTT Finland.
3. Ammonia as a zero-carbon fuel and h2 carrier, **Garcia Serna, Juan (PI)**, 01/11/2024-31/10/2027, PCI2024-155086-2, 175000 €.
4. PID2020-119249RA-I00, SIPHONY - Super-foaming of polymers by combination of sub an supercritical carbon dioxide and water. Ministry of Science and Innovation. **Danilo Cantero (PI)**. (University of Valladolid). 01/09/2021-31/08/2025. 227.480 €.
5. Production of bio-oils from Residual Biomass in Continuous Hydrothermal Reactors. Funding: Agencia Estatal de Investigación; TED2021-129837B-C42. European Union - Next Generation UE; Recovery, Transformation and Resilience Plan; MICINN. Ministry of Science and Innovation; University of Valladolid (UVa) (Program: Ecological Transition 2021). **PI: Cantero Sposetti, Danilo Alberto**. 01/12/2022. 30/11/2024. 184000 €.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Patentes de invención
1. Juan García Serna; Gianluca Gallina. P 201631191. Multi-bed process and pilot plant for biomass fractionation. Spain. 11/12/2017. Universidad de Valladolid.
2. Alvaro Cabeza Sánchez; Juan García Serna; Francisco Sobrón Grañon; Fidel Mato Chaín; Rafael B. Mato Chaín. VA-307-2016. CHEMEN, Process Simulator Suite in Chemical Engineering Spain. 23/09/2016. Universidad de Valladolid
3. Danilo Cantero Sposetti. WO2024010803A1. Cellulose-containing compositions, methods for their preparation, and uses comprising them Estados Unidos de América. 11/01/2024. Renmatix.
4. Danilo Cantero Sposetti. WO2021251987A1. Method of Dispersing Hydrophobic Substances in Aqueous Cleansing System 16/12/2021. Renmatix.
5. Jeremy Austin; Danilo Cantero; Stephen Harris. WO2019195033A1. Composites containing cellulose-based compositions Estados Unidos de América. 10/10/2019. Renmatix.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	none

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

• **Identification of impurities** by GC-MS and complementary techniques, with proven experience on perfluorocarbon liquids (perfluorooctane) in vitreoretinal surgery, in collaboration with the IOBA retina group.
• **Purification and scale-up from lab to pilot plant:** design of reactions and purification routes to obtain high-purity, ophthalmic-grade liquids and materials, and transfer of these processes to pilot scale (TRL 3–5) in PressTech lab, a 500 m² high-pressure facility.
• **Supercritical fluid technology** for formulation and particle engineering: solvent-free extraction, purification, microencapsulation and particle generation (SAS, PGSS, aerogel drying), and recovery of natural antioxidant compounds (e.g. ferulic acid, carotenoids) and biopolymers for potential applications.
• **Process simulation, sustainability and AI**: process modeling, techno-economic analysis, life cycle assessment (LCA) and AI-based tools to evaluate the cost, scalability and environmental footprint of new therapies and products from an early stage.

Capabilities or expertise you would especially like to access through the consortium

• **Knowledge on preclinical testing**, to design materials and products safe and efficient as intended.
• **Clinical perspective**, to better understand the real needs of patients and clinicians and focus our work where it matters most.
• **Regulatory and translational know-how**, to learn how new products move from the lab to clinical use.
• **New fields and ideas**, such as drug delivery, advanced therapies and data science, where chemical engineering can open new opportunities.
• **Long-term collaboration**, through joint projects, funding applications, publications and mobility for our young researchers.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No

X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None.

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X Understood

Name: Juan García Serna
Date completed: 22 of September 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)